

To Veil or Not to Veil? Assessing the Removal of Headscarf Ban in a Muslim Country*

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Abstract

This paper examines the labor and marriage market effects of lifting the headscarf ban in Turkey. Using Household Labor Force Survey data, we employ a difference-in-differences approach, leveraging novel methods to predict veiling status. The reform increased veiled women's employment, primarily in the public sector, while non-veiled women shifted to private-sector jobs. Veiled women also experienced lower informal employment and greater job stability. Marriage rates remained unchanged, but veiled women became more likely to marry wage-earning partners with a larger spousal age gap. Effects were similar across regions with high and low veiling prevalence, suggesting broader implications for minority contexts.

JEL: J16, J12, J22, K31

Keywords: headscarf ban, women's employment, marriage market, identity-based policy

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1 Introduction

Whether the regulation of veiling practices facilitates women’s integration remains a subject of intense debate across many countries.¹ These regulations, often framed around gender equality and secularism, remain highly polarizing. The rationale behind veiling restrictions in public spaces is to promote women’s economic and societal integration by fostering a secular public sphere and reducing social pressures that might limit women’s autonomy, thereby expanding their access to education and employment (Keddie, 1990; Kuru, 2008; Bilge, 2010; Selby, 2011).

Economic theories of identity suggest that individuals face trade-offs between identity and economic opportunity (Akerlof and Kranton, 2000; Oh, 2023). If veiling restrictions impose significant costs on religious expression, some women may opt out of employment rather than compromise their religious identity, while others may adjust their practices when economic incentives outweigh these costs (Carvalho, 2013). Reflecting this theoretical ambiguity, existing empirical evidence on veiling regulations is mixed (Abdelgadir and Fouka, 2020; Maurin and Navarrete H, 2023), highlighting the need for further research on how policies targeting cultural and religious identity shape women’s economic and social choices.

This paper provides new empirical evidence on the labor and marriage market effects of lifting veiling restrictions by analyzing the repeal of the headscarf ban in Turkey. Prior to 2013, wearing headscarves in all public institutions was prohibited in Turkey due to its constitutionally secular structure. In October 2013, the Turkish government lifted the long-standing ban, allowing women to wear headscarves in public spaces, including public institutions and organizations. Given that approximately 60% of Turkish women wear headscarves, this reform presents a unique opportunity to examine how removing institutional barriers related to religious identity affects women’s labor supply and household formation.

Using data from the 2010–2017 Turkish Household Labor Force Survey (HLFS), we employ a difference-in-differences strategy to estimate the effects of the policy change. A key empirical challenge in the literature on headscarf bans is the lack of direct information on veiling status in large-scale surveys, which has often constrained identification strategies. To address this limitation, we leverage auxiliary data from the 2013 Turkish Demographic and Health Survey (DHS), a nationally representative survey uniquely providing individual-level information on veiling status along with extensive demographic and socioeconomic characteristics. This enables us to implement novel prediction methods to accurately estimate veiling probabilities in the HLFS based on observable characteristics. Unlike prior studies that rely on broad proxies (e.g., ethnicity, immigrant status), we leverage individual-level survey data to predict veiling status with greater precision, improving causal identification. Using these predicted probabilities, we implement nearest-neighbor matching (Rosenbaum and Rubin, 1983) to infer veiling status in the HLFS (see Ridder and Moffitt (2007) for a survey of related approaches), enabling a quasi-experimental comparison of labor and marriage market outcomes between veiled and non-veiled women before and after the policy change. We further assess the robustness of our approach using alternative statistical techniques and machine learning algorithms (Mullainathan and Spiess, 2017; Athey and Imbens, 2019; Heller et al., 2024), ensuring that our results are not sensitive to modeling assumptions or sample

¹Throughout the text, we use the term “veiling regulations” to refer to policies governing the wearing of headscarves, turbans, and chadors. These regulations exist across Europe and several Muslim-majority countries. For example, France banned all types of veiling in state-run institutions, such as schools and hospitals, in 2004. More recently, in 2023, the European Court of Justice (ECJ) ruled that EU Member States may prohibit government employees from wearing religious symbols, including headscarves. Similarly, Kazakhstan, where a majority of women practice Islam, has banned headscarves for students and teachers in schools.

selection.

Our findings reveal that following the removal of the headscarf ban, women predicted to be veiled experienced a 1.1 percentage point increase in employment (4% relative to the sample mean) compared to their non-veiled counterparts. This increase is primarily driven by a substantial rise in public sector employment among veiled women, where veiled women’s participation was near zero before the reform. Moreover, we document a notable decrease in the probability of veiled women working as unpaid family workers, suggesting that the removal of the ban also influenced the preferences and employment decisions of women already engaged in the workforce. Consistent with these findings, we observe a decline in the probability of informal employment and an increase in the likelihood of veiled women securing permanent contracts.²

Interestingly, we also find that non-veiled women experienced a shift away from public-sector jobs toward private-sector employment, despite overall public-sector hiring remaining stable. This suggests that veiled women displaced non-veiled women in public employment, highlighting the redistributive labor market effects of identity-based policy changes.

Beyond labor market effects, we examine spillovers into household formation, exploring whether improved labor market opportunities for veiled women influenced partner selection and marriage outcomes. While marriage rates and educational assortative matching remained largely unchanged, with the removal of the headscarf ban, veiled women became 1.3 percentage points more likely to marry wage-earning partners and experienced an increase of approximately 0.8 years in the age gap with their spouses. These patterns suggest that the reform altered marital sorting. One possible explanation is that improved labor market opportunities for veiled women expanded their potential marriage pool, increasing their likelihood of matching with older, wage-earning men who were more established in the labor force. Alternatively, the results may reflect differences in marriage timing if veiled women postponed marriage more than their potential spouses, leading them to match with relatively older and economically stable partners. While our findings are consistent with these interpretations, our data do not allow us to fully disentangle the exact mechanism.

Given that veiling is often associated with lower employment, both in Muslim-majority and Western contexts (Ghumman and Jackson, 2010; Abdelhadi, 2019; Fernández-Reino et al., 2023), our findings contribute to the broader debate whether veiled women are less economically active primarily due to individual religious identity constraints or institutional barriers. By analyzing a large-scale policy change in a Muslim-majority context, we provide robust causal evidence that reducing institutional barriers to religious expression encourages labor market participation and reshapes marriage market dynamics, enriching the understanding of household-level implications of institutional changes. Despite the increasing prevalence of veiling regulations, systematic evidence on their economic and social consequences remains limited and mixed. For example, focusing on the 2004 French headscarf ban in schools, Abdelgadir and Fouka (2020) compare Muslim students to native students and find that exposure to the ban significantly reduced secondary educational attainment among female students with North-African origin, with long-term negative effects on their labor market outcomes and family composition. Taking a similar empirical approach but focusing on a different cohort, Maurin and Navarrete H (2023) analyze the 1994 ministerial circular instructing school administrators to prohibit veiling in schools and find that this earlier ban improved educational outcomes for female students with a Muslim background.

While previous studies primarily examine the effects of imposing veiling bans, our paper

²Analyzing the occupational distribution within the public sector, the largest effects are observed in technician and associate professional roles, such as healthcare assistants and administrative secretaries, which typically require lower levels of educational attainment.

provides one of the first causal estimates on the consequences of lifting such a restriction. Unlike in Western contexts, where veiled women typically constitute a minority—often immigrants—in Turkey, veiled women represent the majority of the female population. Given these key differences in the treated populations, we conduct a heterogeneity analysis comparing veiled women in regions with high and low prevalence of veiling to understand whether the effects of the reform vary by social context. We find no significant differences in labor market responses between these groups, suggesting that peer effects and local social acceptance do not appear to be the primary drivers behind these outcomes. This result carries broader implications for contexts such as France, where veiled women form a minority and may experience greater discrimination through mechanisms like Islamophobia. As veiled women in Turkey respond similarly irrespective of local veiling prevalence, lifting a headscarf ban could yield comparable effects even in societies where veiling is relatively uncommon and negatively perceived. Moreover, our findings encourage a reconsideration of how integration is conceptualized. In contexts like Turkey, integration may not necessitate assimilation into secular norms; instead, it may simply entail enabling women’s economic and social participation without requiring them to abandon their religious identity. From this perspective, removing identity-based restrictions may promote economic inclusion independently of cultural assimilation.

In the context of Turkey, [Uğur \(2020\)](#) examines the impact of the 1997 headscarf ban on women’s educational attainment and finds no significant effect on university completion, largely due to already low enrollment among veiled women. More closely related to our analysis, [Corekcioglu \(2021\)](#) examines the 2013 removal of the headscarf ban using a difference-in-discontinuities design, focusing exclusively on municipal employment outcomes determined by closely contested mayoral elections. Consistent with our findings, she shows an increase in female employment, but only within municipalities governed by Islamist mayors. Our study significantly extends this evidence by analyzing the labor market as a whole—covering both public and private sector employment—and exploring effects on household formation decisions, thus providing a comprehensive picture of the policy’s broader societal impact. Although our results confirm that municipalities with Islamist mayors experienced stronger effects, we clearly show that the removal of the ban influenced women’s economic outcomes, irrespective of political affiliation or local governance.

More broadly, our study contributes to research on how legal and institutional barriers shape women’s labor market participation. Recent literature highlights the role of discriminatory laws in limiting women’s economic opportunities globally, demonstrating that legal reforms promoting gender equality can significantly boost female employment rates ([Roy, 2019](#); [Hyland et al., 2020](#)). Studies of specific institutional reforms also show that removing discriminatory constraints, such as restrictions on women’s work hours or barriers in family law, leads to increased female participation in the labor market ([Hallward-Driemeier and Gajigo, 2015](#); [Gonzales et al., 2015](#)). By framing the headscarf ban as a gender-discriminatory policy, our study extends this literature, demonstrating how removing legal constraints related to religious identity affects both labor market participation and household formation decisions of women.

Finally, our study contributes to the literature on traditional gender norms as barriers to female employment, particularly in developing countries. [Jayachandran \(2021\)](#) emphasizes that restrictive norms concerning women’s mobility, household responsibilities, and community expectations significantly reduce female labor force participation. Prior studies have shown limited impacts of economic empowerment policies—such as vocational training or increased control over earnings—due to deeply entrenched gender norms ([Field et al., 2010](#);

Duflo, 2012; Field et al., 2021; Gazeaud et al., 2023). Although veiled women in Turkey generally adhere to conservative gender norms, typically linked to lower employment rates (Dildar, 2015), we find that removing the institutional barrier to veiling significantly increased their employment, especially within the public sector. Our findings thus suggest that institutional constraints may substantially outweigh traditional norms in determining women’s economic choices, challenging assumptions that conservative norms alone drive low labor force participation among veiled women.

The remainder of the paper is structured as follows: Section 2 discusses the the headscarf debate and public sector employment in Turkey. Section 3 describes the data and empirical strategy. Section 4 presents the main findings and alternative analyses. Section 5 concludes.

2 Institutional Background

Historical and social context of veiling regulations. Turkey, a constitutionally secular country since the 1924 Constitution, has a predominantly Muslim population. The wearing of headscarves in public has long been a contentious issue, contributing to tensions between secularism and religious freedoms. The historical evolution of headscarf bans in Turkey is examined in detail by Cindoglu and Zencirci (2008). While precise data on the prevalence of veiling are limited, multiple sources provide estimates over time. According to the report by KONDA (2019), approximately 97% of Turkey’s population identifies as Muslim. Figure 1 illustrates trends in the proportion of women wearing headscarves in Turkey from 2010 to 2019, based on data from these sources. The percentage of veiled women was approximately 66% in 2010, decreasing to 62% in 2011. This figure remained stable until 2013, after which a slight increase occurred (around 2%). Similarly, data from the 2008 and 2013 waves of the Turkish Demographic and Health Survey (DHS) reflect comparable trends: 72% of women wore headscarves in 2008, dropping to 62% in 2013.³

Following the 1980 military coup, the first formal headscarf ban was enacted in 1982, which prohibited headscarves in public spaces. This regulation sought to reinforce secularism, a foundational principle of the Turkish Republic. However, the ban sparked significant societal discord, especially among conservative groups who viewed it as an infringement on religious freedoms and emblematic of secularist authoritarianism. In 1997, the government extended the ban from universities to preparatory schools (Uğur, 2020).

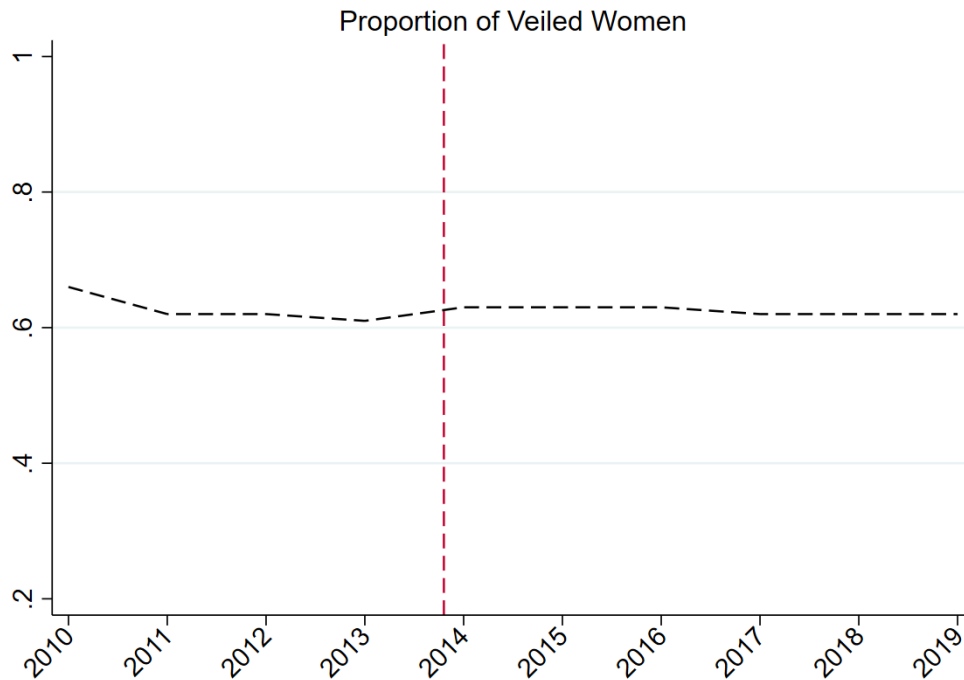
In 2008, the Turkish government made its first significant effort to address the longstanding headscarf ban. This initiative aimed to lift the ban in universities by amending Articles 10 and 42 of the Turkish Constitution. However, the Constitutional Court intervened, ruling on June 5, 2008, that the proposed amendments violated the preamble of the Constitution and the principle of secularism enshrined within it. The Court subsequently annulled the proposed changes, effectively blocking their implementation (Wiltse, 2008; Höjelid, 2010). Despite this setback, the Higher Education Council partially lifted the headscarf ban in 2010 by revising regulations within its jurisdiction. These regulatory changes marked a gradual easing of restrictions on veiling in universities across Turkey (Bianet, 2010; BBC News, 2010).⁴

A landmark change occurred on October 8, 2013, when the government, under Prime Minister Erdogan, implemented a “democratization package”. This reform package aimed to address various human rights and political issues, including improvements in minority

³Note that the 2008 DHS includes data on ever-married women, while the 2013 DHS collects data on all women.

⁴In the following section, we address its potential implications for our results.

Figure 1: Share of Veiled Women Between 2010-2019 in Turkey



Notes: Data sourced from KONDA survey data ([KONDA, 2019](#)).

rights and democratic governance, such as allowing the use of Kurdish in public schools and enabling the establishment of educational institutions where Kurdish could be taught. While the central focus of this reform package was to expand the rights of minorities, one of its most notable aspects was the dismantling of the longstanding headscarf ban within public spaces. The reform officially came into effect on October 8, 2013, repealing the 1982 by-law that prohibited headscarves.

Prior to this policy change, veiled women were formally prohibited from working in public institutions while wearing a headscarf. Although a small number of veiled women managed to circumvent the ban by wearing wigs in the workplace, this practice remained limited and was not officially recognized by authorities ([Cindoğlu, 2011](#); [Guveli, 2011](#)). With the reform, women were legally permitted to wear headscarves in public institutions and government offices, with exceptions for personnel in the armed forces, security forces, and judiciary ([Akoglu, 2015](#); [Karahan and Tuğsuz, 2022](#)). The ban was subsequently lifted for members of the judiciary in 2015 and for security forces in 2016.

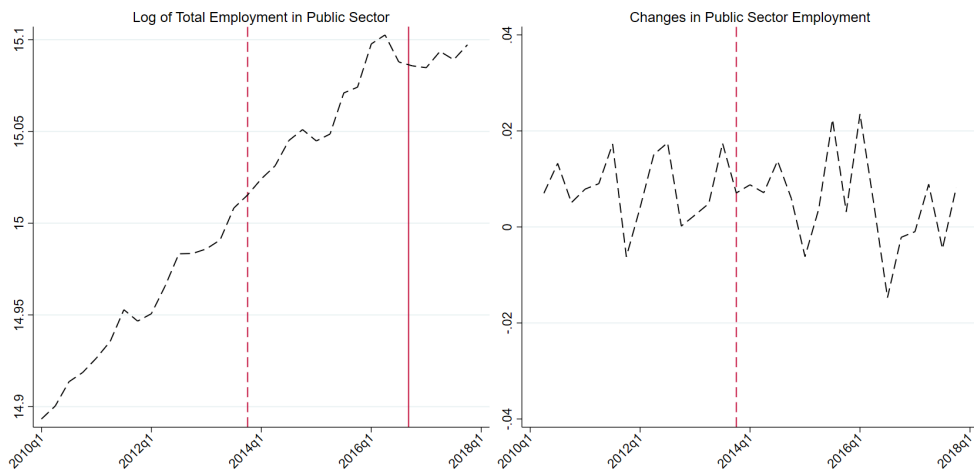
Given the democratization package's primary focus on political rights and the short time between its announcement and implementation, it is plausible to assume that the removal of the headscarf ban likely came as an unexpected change. Subsequent sections provide evidence to support this assumption through an analysis of pre-policy trends.

Public sector employment. Public sector employment in Turkey accounts for approximately 13% of total employment, below the OECD average of 18% in 2019 ([OECD, 2019](#)). Figure 2 presents trends in total public sector employment in Turkey between 2010 and 2018, based on quarterly data from the Turkish Presidency of Strategy and Budget. The left-hand graph shows the logarithmic growth in public sector employment, which rose steadily each year from 2010 to 2016. However, this upward trend paused during the second quarter of 2016, likely due to the military coup attempt in July 2016, which led the central government to freeze public sector hiring between the second and fourth quarters of that year due to security concerns and government purges ([BBC News, 2017](#); [Euronews, 2017](#)). Public sector employment followed a moderate upward trajectory after this period. The right-hand graph of Figure 2 suggests that the labor demand side of the public sector remained stable from

2010 to 2017, with no major disruptions apart from the temporary hiring freeze in 2016.

Public sector recruitment in Turkey is centralized and highly standardized. Applicants must pass a civil service examination administered biennially by the Department of Measuring, Selection, and Placement (OSYM). This exam targets different groups of candidates in even and odd years. Odd-year exams, which primarily focus on positions such as teachers and district governorates, cover a smaller proportion of candidates. In contrast, even-year exams attract a broader pool of applicants seeking positions such as engineers, teachers, technicians, technical staff, and clerks. Figure 3 illustrates the proportion of individuals taking the public sector entrance exam during even years from 2008 to 2018. Notably, the lack of a significant pre-reform increase in exam participation supports the assumption that the removal of the headscarf ban was largely unanticipated. Instead, the exam participation surged in 2016, likely reflecting increased interest in public sector employment

Figure 2: Employment in public sector as a percentage of total employment



Notes: Data sourced from the Turkish Presidency of Strategy and Budget.

Public sector positions are predominantly professional roles, such as teaching, nursing, and engineering, which require advanced qualifications. Women constitute 57% of these roles, whereas men hold 41%. Clerical positions, which require lower educational qualifications, employ 17% of women and 15% of men in the public sector. Technical roles, involving specialized tasks, are the third most common for both genders. By contrast, labor-intensive roles—such as agricultural and craft work—are male-dominated, accounting for 28% of male employment versus 9% of female employment. Positions as legislators and senior officials remain rare, comprising 4% of male and 3% of female employment. Detailed gender breakdowns across public sector occupations are provided in Table A1 in the Appendix.

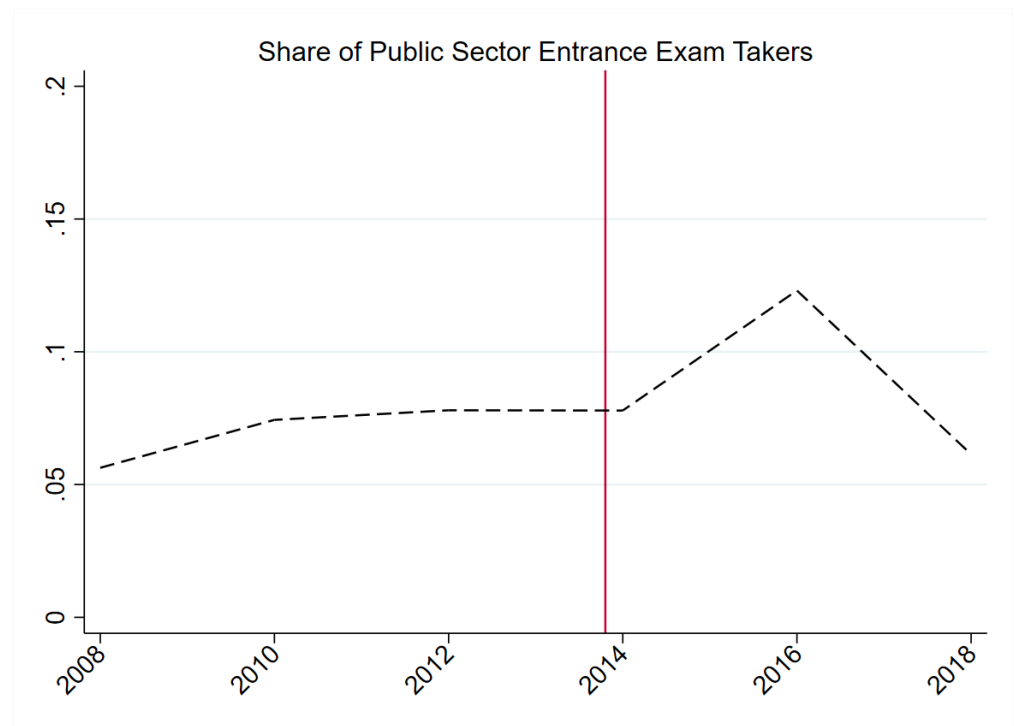
3 Data and Empirical Strategy

3.1 Data

Turkish Household Labor Force Survey. The primary dataset for this study is the Turkish Household Labor Force Survey (HLFS), an annual survey conducted by the Turkish Statistical Institute. The HLFS is nationally representative, covering approximately 150,000 households per wave, and provides detailed demographic and labor market information for individuals aged 16 and above. For this analysis, we use repeated cross-sections spanning 2010 to 2017.⁵

⁵Data from 2018 onward are excluded to avoid confounding effects from the July 2016 military coup attempt and subsequent mass public sector dismissals to ensure that observed changes are driven by the

Figure 3: Proportion of individuals who take public sector entrance exam



Notes: Data sourced from the Department of Measuring, Selection, and Placement.

The HLFS includes a wide array of variables, including individual characteristics (e.g., age, marital status, education, household size, and region of residence) and detailed labor market information, such as employment type and status (e.g., public or private sector, self-employment or unpaid family work), occupation, weekly hours worked, and monthly earnings. Earnings data incorporate bonuses and premiums received during the reference month.⁶ This comprehensive dataset enables a detailed analysis of the labor market effects of the reform.

The sample focuses on women aged 18–49, who are most likely to be active in the labor market. A limitation of the HLFS is that earnings data are only available for wage earners. Therefore, when we analyze the impact on earnings, we are only able to provide results for those who work as wage earners. In total, our sample of analysis consists of 935,476 women. Column 1 of Table A2 in the Appendix presents descriptive statistics for this sample, detailed background characteristics, labor market outcomes and partner characteristics (conditional on being married). Panel A shows that the average age of women in the sample is approximately 33 years old, with 98% being native. About 35% have at least completed high school, while 15% hold a university degree. Notably, around half of the women in the sample have only primary education or no formal educational attainment. Moreover, on average 71% of women are married in the sample and the average household size is about 4, as indicated in the last row of Panel A. Panel B summarizes labor market outcomes. About 33% of women are employed, primarily as wage earners (22%), including 5% in the public sector. Moreover, 8% are unpaid family workers, and 3% are self-employed. Panel C provides partner and household characteristics for married women. The average age of partners is 39.6 years, with an average age gap of 4 years between spouses. Around 16% of partners hold a university degree, and 45.9% have at least completed high school. Employment rates among partners are high, with 83% employed and 60% working as wage earners.

Turkish Demographic and Health Survey. While the HLFS provides detailed labor

removal of the headscarf ban rather than broader political and economic disruptions.

⁶Monthly wages are deflated using the Consumer Price Index (CPI) and converted to 2017 Turkish Liras, then expressed in U.S. dollars using 2017 exchange rates.

market data, the 2013 Turkish Demographic and Health Survey (DHS) offers a unique data on headscarf-wearing practices, allowing us to distinguish between veiled and non-veiled women. The DHS, conducted by Hacettepe University’s Institute of Population Studies, provides nationally representative data for 9,746 women, 6,835 of whom were married at the time of the survey (conducted from September to December 2013). In addition to information on headscarf-wearing practices, The DHS provides detailed socioeconomic and demographic data.

To complement the HLFS, we use the 2013 Turkish Demographic and Health Survey (DHS), conducted by Hacettepe University’s Institute of Population Studies. The DHS provides nationally representative data for 9,746 women, 6,835 of whom were married at the time of the survey (conducted from September to December 2013). The DHS offers unique information on headscarf-wearing practices, a key variable for this analysis, with detailed socioeconomic and demographic data.

In the 2013 DHS, 62% of women aged 18-49 reported wearing a headscarf. Table A3 in the Appendix compares non-veiled (Column 1) and veiled (Column 2) women. Non-veiled women are significantly more likely to have completed high school or university, whereas veiled women predominantly have primary education or less.⁷ Labor market outcomes (Panel B) indicate that non-veiled women are more likely to be employed and work as wage earners. Before the reform, only 1% of veiled women worked in the public sector, compared to 10% of non-veiled women, illustrating the impact of prior restrictions.⁸ Panel C highlights partner characteristics. Husbands of non-veiled women are, on average, one year younger than those of veiled women and more likely to hold a university degree or high school diploma.⁹ Moreover, veiled women tend to live in larger households compared to their non-veiled counterparts.

3.2 Empirical strategy

The removal of the headscarf ban directly affects veiled women, making it plausible to assume that they experience a greater impact from this policy change than their non-veiled counterparts. Given that the majority of women in Turkey wear a headscarf, a straightforward approach to identify the effects of the policy is to compare labor market outcomes of all women before and after the removal of the ban. Assuming that all women in the sample wear a headscarf, we estimate the following equation:

$$y_{it} = \theta_1 + \theta_2 Post_t + \eta_i + \mu_{it}, \quad (1)$$

where y_{it} represents the outcome of interest for woman i at time t . The dummy variable $Post_t$ takes a value of 1 for the post-policy period ($t \geq 2014$) and 0 for the pre-policy period ($t < 2014$). The coefficient θ_2 captures the average difference in labor market outcomes for women before and after the policy change.

While this specification serves as a useful benchmark, providing an upper-bound estimate of the policy’s potential effect, assuming all women as veiled might introduce misclassification, as not all women were directly affected by the reform. This misclassification likely

⁷Among non-veiled women, 39% have a university degree compared to 6% of veiled women. Conversely, 65% of veiled women have primary education or less, compared to 20% of non-veiled women.

⁸The presence of veiled women in public-sector employment before the reform likely reflects cases where women circumvented the ban by wearing wigs in the workplace, as discussed in the institutional background section.

⁹Approximately 32% of non-veiled women’s partners have a university degree, compared to 10% for veiled women. Similarly, 58% of non-veiled women’s partners have a high school diploma, compared to 28% for veiled women.

attenuates the estimated effect, resulting in a noisier estimate of the true impact. To better isolate the treatment effect, we leverage individual-level veiling data from the 2013 Turkish Demographic and Health Survey (DHS), which allows us to estimate veiling probabilities rather than relying on broad proxies, such as Muslim-majority immigrant status (Maurin and Navarrete H, 2023; Abdelgadir and Fouka, 2020), as we detail below. This approach enables us to define a treatment group of women more affected by the policy and a comparison group of women who were less or not affected.

Specifically, we define treatment status based on predicted veiling probabilities and construct a binary indicator, *Veiled*, which equals 1 if woman i is predicted to wear a headscarf and 0 otherwise. Using this definition, we estimate the policy effect in a difference-in-differences (DiD) framework, comparing changes in labor and marriage market outcomes for predicted veiled and non-veiled women before and after the reform. Formally, we estimate the following specification:

$$y_{it} = \beta_1 + \beta_2 (Veiled_i * Post_t) + \beta_3 (Veiled_i) + \eta_i + \mu_t + \varepsilon_{it}, \quad (2)$$

where μ_t represents year fixed effects. The estimated coefficient on the interaction term, β_2 , is the difference-in-differences estimator, capturing the average treatment effect between treated and non-treated women following the policy change.

In each specification, the vector η_i includes Dummy variables for educational attainment (no formal education, primary school, junior high school, high school, university, or higher, age, age squared, a dummy variable indicating whether the woman is a native, and dummies for each region of residence at the NUTS-2 level. We also include interactions of region dummies with year fixed effects to control for time-varying regional heterogeneity.

Out-of-sample predictions. Our empirical strategy, as outlined in Equation (2), hinges on defining treatment and control groups based on exposure to the removal of the headscarf ban, which we infer through veiling status. Ideally, this status would be directly observable in the main dataset. However, since the HLFS does not collect information on veiling, we utilize a propensity score matching (PSM) technique introduced by Rosenbaum and Rubin (1983) to predict veiling status using a unique dataset from the DHS. This approach, adapted from its traditional use in balancing covariates for causal inference, allows us to predict treatment status in the HLFS using external data from the DHS.

It is a two-stage procedure. First, using data from the DHS, we estimate the propensity scores of wearing a headscarf by fitting a Probit model. The dependent variable is binary, taking the value of one if the woman wears a headscarf and zero otherwise. The regressors include dummy variables for the woman’s education levels, a dummy variable indicating whether the woman is native, age and region dummies at the NUTS2 level, and household size.¹⁰ Table 1 presents the estimated regression results and corresponding marginal effects of each variable on the probability of wearing a headscarf. Using these covariates, we predict the propensity scores for wearing a headscarf for both the DHS and HLFS samples. While the model fit could potentially be improved with additional control variables, the HLFS dataset unfortunately lacks further relevant information.

The distribution of predicted probabilities is illustrated in Figure 4. The left graph displays the distribution of predicted probabilities for wearing a headscarf using the DHS sample, while the right graph displays the probability distribution of women in the HLFS. The substantial overlap in the distributions of predicted probabilities in both samples suggests that for each woman in the HLFS, there is a comparable female respondent in the DHS

¹⁰Marital status is excluded to avoid potential endogeneity, as it is analyzed as an outcome variable in later sections. Nevertheless, including marital status in the prediction model does not alter our results.

based on observed covariates.

In the second stage, we use the predicted propensity scores to perform a nearest-neighbor matching based on these covariates.¹¹ We then impute veiling information (yes = 1, no = 0) for each woman in the HLFS using the veiling status of their nearest neighbors from the DHS dataset.¹² In the matched HLFS sample, approximately 61% of women are predicted to wear headscarves, with a standard deviation of 0.48, closely mirroring the DHS sample (62% with a standard deviation of 0.48). This similarity in mean values suggests a high degree of comparability between the two samples. Descriptive statistics for key variables by veiling status in the estimation sample are presented in columns 2 and 3 of Table A2 in the Appendix.

Table 1: Probit Estimates of Probability of Wearing a Headscarf

Variables	Coef. (1)	Std. Err. (2)	Marginal Eff. (3)
Age	-0.010***	0.003	-0.002***
Native	0.626***	0.135	0.158***
Household size	0.168***	0.017	0.042***
Education level:			
Primary school degree	-0.337***	0.097	-0.068***
Junior high school degree	-0.771***	0.111	-0.186***
High school degree	-1.253***	0.106	-0.347***
University degree	-2.010***	0.114	-0.600***
Obs.	7,172		
Pseudo R ²	0.279		

Notes: Data are from the 2013 Turkish Demographic and Health Survey. The sample includes all women aged 18-49. The estimates include dummies for each region at the NUTS2 level. The omitted variable for education level is having no education. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

We acknowledge potential concerns with predicting veiling status from external data, particularly regarding the conditional independence assumption underlying this prediction. To address these concerns and further validate our results, we employ alternative robustness checks, including statistical matching methods allowing for conditional correlations and alternative predictions using machine learning techniques. These robustness analyses, along with detailed placebo and sensitivity tests, are presented in detail in the following section.

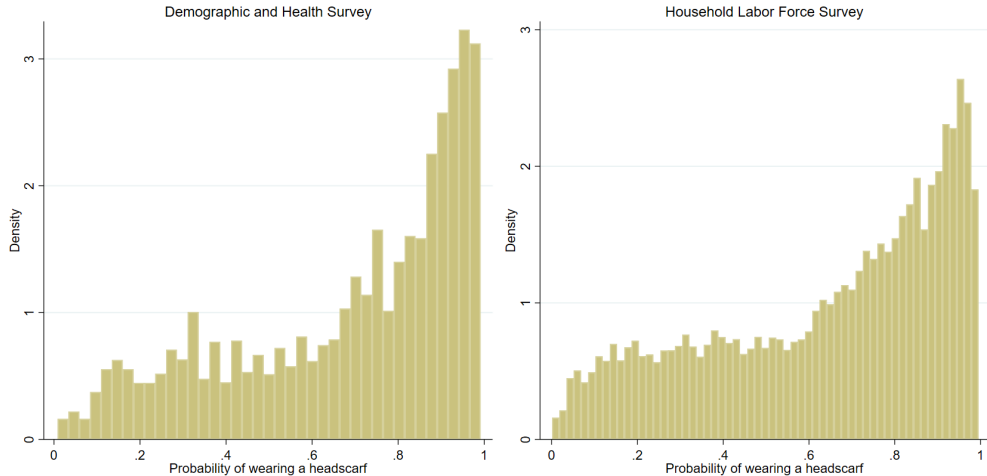
Another key consideration is whether the predicted treatment and control groups are comparable based on observable characteristics. To address this, in each estimate, we include a set of covariates η_i that control for potential differences, such as age, education level, household size, and whether the woman is native. We also control for region and year fixed effects, with their interactions, to account for any region-specific time variations that could affect veiled and non-veiled women differently.

Moreover, the assignment of pre- and post-treatment periods could potentially influence our results. Since the Turkish government lifted the headscarf ban for women in October 2013, we designate 2014 as the start of the post-treatment period in our main analysis, given

¹¹To perform the nearest-neighbor matching algorithm, we use psmatch2 command in Stata.

¹²We use matching with replacement. Specifically, each observation in the DHS can be used as a match for multiple observations in the HLFS. Additionally, we set the maximum distance between two propensity scores to 0.05. If the propensity distance exceeds this threshold, no observation is matched and we drop this observation (There are 1,416 observations in total, which is less than 1% of the estimation sample).

Figure 4: Distribution of probabilities of wearing a headscarf



Notes: The left-hand graph shows the distribution of predicted probabilities of veiling based on data from the DHS, while the right-hand graph presents the distribution using data from the HLFS.

the annual nature of the data.¹³ To assess the sensitivity of our results to this timing, we also re-estimate our main specifications by excluding the 2013 wave.

Another concern pertains to the potential inclusion of women affected by other policy changes in our main estimation, which could complicate the identification of the actual effects of lifting the headscarf ban. As discussed in the institutional background section, the gradual lifting of the headscarf ban in universities from 2010 onward may have influenced women’s labor market outcomes, particularly if it altered their decisions to pursue higher education. Moreover, the well-documented effects of the 1997 compulsory schooling reform on women’s educational and labor market outcomes further add to this complexity (Erten and Keskin, 2018; Güneş, 2016; Merlino and Yurdakul, 2024).¹⁴ To address these concerns, we include dummy variables in our estimation to separately account for whether a woman was affected by the removal of the headscarf ban in universities in 2010 or by the 1997 compulsory schooling reform. These variables are constructed using birth year information. Using the birth cohort differences allows us to disentangle the effects of these earlier policy changes from the impact of lifting the broader headscarf ban in 2013.

Finally, we tackle concerns regarding potential selection bias in imputing veiling status. In our main analysis, we estimated veiling probabilities using the 2013 DHS data. While we do not expect significant changes in veiling probabilities immediately after the policy change, relying on the 2013 data could introduce bias if the policy influenced veiling choices (see 1). To mitigate this, we re-estimate veiling probabilities using imputation data from an earlier wave, prior to the 2014 policy change. For our sensitivity checks, we choose the 2008 DHS wave as the reference.

¹³Although the Turkish Statistical Institute collects data on a monthly basis, the specific survey month is not provided.

¹⁴In Turkey, two major reforms have extended the duration of compulsory education: (i) the 1997 reform, which mandated that individuals born after 1986 complete junior high school (8 years of education), meaning the oldest individuals affected were 30 years old in our sample; and (ii) the 2012 reform, which requires individuals born after 1997 to complete high school (12 years of education). Notably, since the oldest individuals affected by the 2012 reform were only 19 years old by 2017, a relatively small proportion of women in our sample were impacted by this later reform.

4 Main Results

4.1 Effects on Labor Market Outcomes of Women

We begin by examining trends in key labor market outcomes for women predicted to be veiled and non-veiled between 2010 and 2017. Figure 5 provides an initial look at potential changes in women’s employment, wage employment (public and private sector employment), and self-employment during this period. The raw data reveal no significant pre-policy differences, with both groups experiencing steady increases in employment probability. Public sector employment for veiled women was close to zero before the policy change but increased after the reform, while private sector and self-employment trends remained stable between both groups.

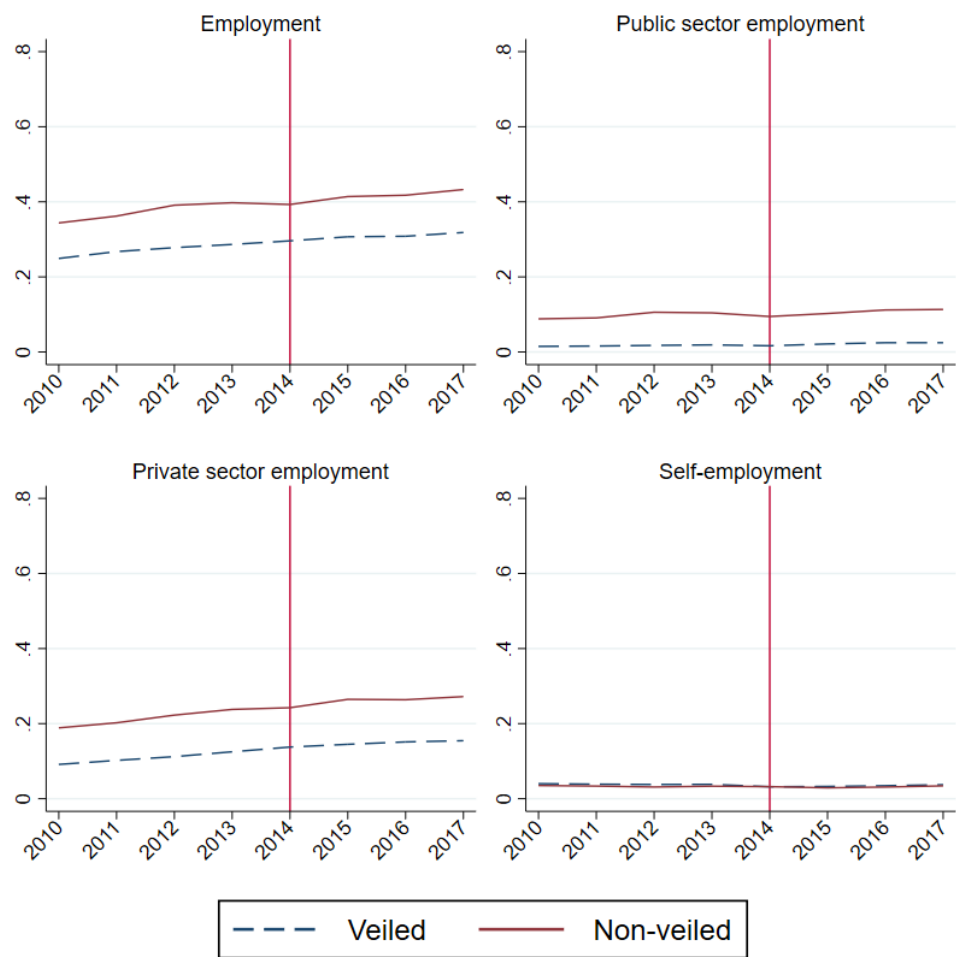
Before turning to the corresponding regression-based estimates, Panel A of Table 2 reports estimates from a simple before-after analysis, focusing on the variable *Post* in Eq. (1). The estimate in Column 1 indicates a 1.9 percentage point increase in the probability of employment for women during the post-policy period compared to the pre-policy period, representing an approximate 6% rise relative to the baseline employment rate of 31%. Column 2 examines public sector employment, showing a 0.6 percentage point decline, which corresponds to a 12% reduction relative to the pre-policy sample mean. As public sector employment was near to zero before the policy change, (see Table A3 in the Appendix) this decline suggest reduced opportunities for non-veiled women in the post-policy. In contrast, Column 3 highlights a 3.3 percentage point increase in private sector employment, representing a 22% rise relative to the baseline. Columns 4 and 5 show a statistically significant decrease of 0.02 percentage points in self-employment and a statistically significant decrease of 0.2 percentage points in unpaid family work.

These results provide an initial overview of how women’s employment patterns evolved following the removal of the headscarf ban. However, these before-after comparisons, while informative under the assumption that all women were affected by the policy change, rely solely on changes over time associated with the implementation of the reform. They do not account for confounding factors such as macroeconomic conditions or concurrent policies, which may have also influenced labor market outcomes during the post-policy period. This limitation is particularly important for outcomes like employment, which are sensitive to broader economic fluctuations. To address these concerns and better isolate the effects of lifting the headscarf ban, we refine our analysis by comparing labor market outcomes between women predicted to be veiled and those predicted not to be veiled in the post-policy period.

Panel B of Table 2 presents the main difference-in-differences estimates evaluating the impact on labor market outcomes, focusing on the interaction term $Veiled \times Post$ in Eq. (2). Column 1 of Panel B indicates a 1.1 percentage point increase in employment among veiled women relative to their non-veiled counterparts following the policy change. This effect corresponds to a 4% increase relative to the average pre-policy employment rate for veiled women, as shown in the last row of Panel B. Columns 2 through 5 of Panel B explore the effects of lifting the headscarf ban on women’s employment across working as wage earners¹⁵, working as self-employed and as unpaid family workers. The estimate in Column 2 reveals a 1.5 percentage point increase in public sector employment among predicted veiled women relative to their non-veiled counterparts, representing a substantial improvement given the near-zero mean of public sector employment for veiled women prior to the reform. In Column 3, the estimated impact on private sector employment is small and statistically insignificant,

¹⁵The wage earners are employees in the public and private sectors, as wages are only reported for these sectors.

Figure 5: Trends in Key Labor Market Outcomes by Veiled and Non-veiled Women



Notes: Data from the 2010-2017 HLFS. Employment is coded as 1 if the woman is employed, and 0 otherwise. Public sector, private sector, and self-employment outcomes are coded as 1 if the woman is employed in the respective sector, and 0 otherwise.

suggesting that the reform had no measurable effect on private sector employment for veiled women compared to non-veiled women. Similarly, Column 4 shows no significant change in the likelihood of self-employment. However, the estimate in Column 5 indicates that the removal of the headscarf ban is associated with a significant 1.1 percentage point decrease in the probability of being an unpaid family worker among veiled women compared to their non-veiled counterparts, implying some significant shifts in labor market behaviors of veiled women who were already employed prior to the reform.

To further evaluate the timing and persistence of these effects, we present event study estimates by plotting the coefficients of interactions between the treatment variable and year dummies. Figure 6 illustrates year-by-year effects of the reform, with corresponding regression results provided in Table A4 in the Appendix. Focusing on the pre-reform period, the estimated coefficients in the top-left graph show no significant changes in employment for predicted veiled women compared to predicted non-veiled women, indicating parallel trends prior to the policy change. Following the reform, however, employment among veiled women significantly increased relative to their non-veiled counterparts, particularly in 2014, 2016, and 2017. More importantly, public sector employment among veiled women shows a substantial and consistent increase after the policy change, as seen in the top-right graph. This upward trend begins immediately after the policy implementation in October 2013 and continues through 2017. Although smaller in magnitude, we also observe a relative increase in public sector employment for veiled women in 2013, likely reflecting the immediate effects of the policy’s announcement in October 2013. By contrast, the bottom graphs of Figure

Table 2: *Effects of the Headscarf Ban Removal on Women’s Labor Market Outcomes*

Outcome	Employment probability (1)	Public sector em- ployment (2)	Private sector em- ployment (3)	Self- employed (4)	Unpaid family worker (5)
Panel A: Before-After Estimates					
Post	0.019*** (0.001)	-0.006*** (0.000)	0.033*** (0.001)	-0.002*** (0.000)	-0.011*** (0.001)
Year Fixed Effects	No	No	No	No	No
Observations	934,060	934,060	934,060	934,060	934,060
Mean Dependent Var.	0.31	0.05	0.15	0.04	0.09
Panel B: Difference-in-Differences Estimates					
Veiled $\times$ Post	0.011*** (0.002)	0.015*** (0.001)	0.001 (0.002)	-0.000 (0.001)	-0.011*** (0.001)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	934,060	934,060	934,060	934,060	934,060
Mean Dependent Var.	0.28	0.02	0.11	0.04	0.11

Notes: Data are from the 2010-2017 HLFS. The sample includes all women aged 18-49. Panel A presents before-after estimates for all women, while Panel B reports difference-in-differences estimates comparing veiled and non-veiled women. The last row in each panel reports the mean outcome for the corresponding group before the policy change. Regressions control for native status, education level dummies, age, age squared, household size, and region and year fixed effects, along with their interactions. Standard errors (in parentheses) are clustered at the individual level. All estimates are weighted using cross-sectional survey weights from the wave in which the outcome was measured. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

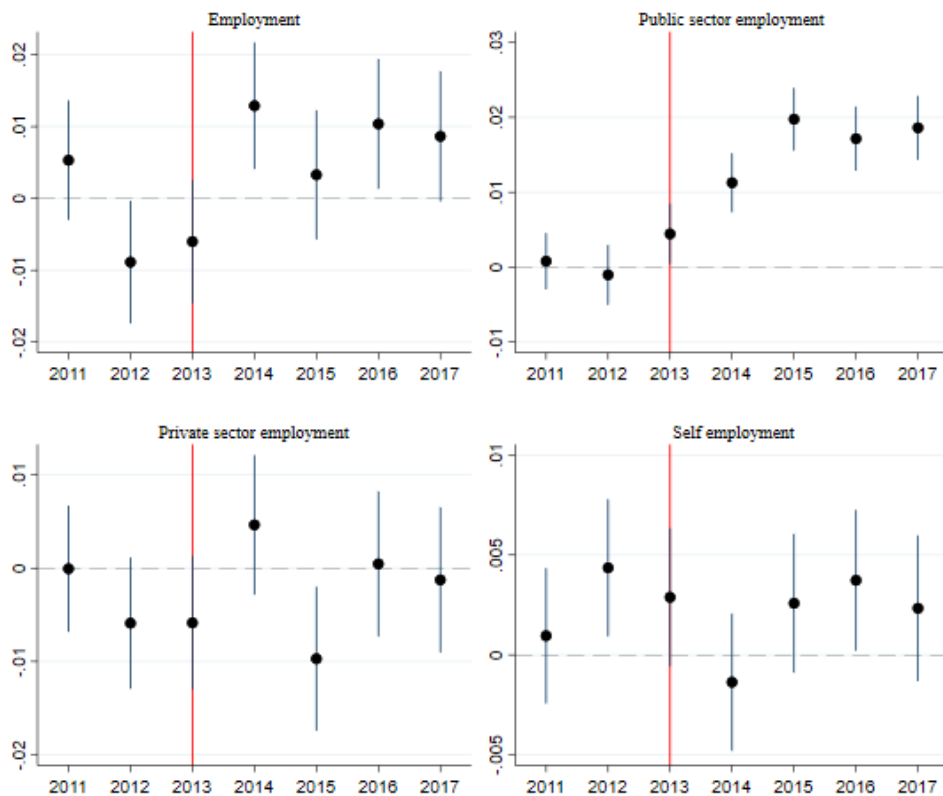
6 show no significant changes in private sector employment or self-employment for veiled women compared to non-veiled women, consistent with the findings in Panel B of Table 2.

Overall, these findings imply that while there were no significant shifts in labor market outcomes during the pre-policy period, lifting the headscarf ban in public institutions led to a significant increase in employment among veiled women, primarily resulted in an increase in public sector employment among these women. Notably, the rise in public sector employment is not solely driven by higher overall employment rates but also by a substantial decline in unpaid family work among veiled women following the policy change. This shift suggests that some veiled women already participating in the labor market before the policy transitioned to public sector roles, likely motivated by the improved opportunities and working conditions made accessible through the removal of the headscarf ban.

To provide evidence supporting this mechanism behind the transition among veiled women who were already employed, we examine the effects of the policy change on informal employment and working with permanent contracts. Estimates in Table A5 indicate that after the removal of the headscarf ban, veiled women were 0.6 percentage points less likely to work under informal contracts and 1.8 percentage points more likely to secure permanent contracts compared to their non-veiled counterparts. These results support the interpretation that some veiled women, already employed prior to the reform, transitioned into the public sector in pursuit of “better” jobs following the removal of the headscarf ban.

In columns 3-5 of Table A5, we also analyze shifts in occupational choices within the public sector, focusing on three main occupations that account for approximately 90% of all female occupations in the public sector in Turkey (see Table A1). Column 3 presents

Figure 6: Event Study Estimates for The Effects of The Policy Change on The Labor Market Outcomes of Women



Notes: Data are from the HLFS. OLS coefficient estimates and their 95% confidence intervals are reported. The controls include whether the woman is a native, dummies for each education level, age, age squared, household size, and region and year fixed effects, with their interactions. Standard errors are clustered at individual level. Data are weighted using the cross-sectional weights for the wave at which the outcome was measured.

estimates for professional roles, which generally require higher education qualifications, such as teachers and healthcare professionals. In contrast, Columns 4 and 5 capture occupations that typically require lower educational attainment, including technicians, associate professionals, and clerical workers. These roles encompass clinical assistants, assistant nurses and midwives, administrative secretaries, and office clerks. While we observe significant increases in employment for veiled women across all occupational categories, these increases are predominantly concentrated in technician and associate professional positions. The probability of veiled women working as technicians and associate professionals rose by approximately 0.8 percentage points, compared to 0.2 and 0.3 percentage point increases for professional and clerical roles, respectively. These findings suggest that the rise in public sector employment among veiled women following the policy are mainly influenced by women with lower levels of educational attainment.

In the last two columns of Table A5, we report the effects on weekly working hours (including unemployed women with zero hours) and hourly earnings.¹⁶ The estimates indicate that lifting the headscarf ban increased veiled women's weekly working hours by around 0.42 hours, representing a 4% increase relative to non-veiled women. Compared to the observed impact on the probability of employment, this effect is primarily driven by higher employment probability rather than a shift in preferences for longer hours.¹⁷ Turning to hourly earnings, we observe some negative effect, which is statistically significant at the 10% level.

¹⁶The analysis for labor market earnings is restricted to wage earners due to data limitations.

¹⁷At this point, one might wonder whether the policy change influenced preferences for longer working hours. However, due to selection into employment post-reform, estimating this effect for the selected sample does not allow for a causal interpretation, as it may lead to inconsistent estimates due to selection bias.

However, given that earnings data are only available for wage earners, this result should be interpreted with caution, as it may reflect changes in the composition of employed women rather than a direct effect of the reform.

Finally, we look at evaluation of the labor market outcomes for women who are predicted to be non-veiled by incorporating the post-policy period dummy variable, *Post*, in Eq. (2) in place of year fixed effects. The results are reported in Table A6 in the Appendix. The estimates suggest that veiled women may have served as substitutes for non-veiled women in the public sector following the removal of the headscarf ban. Specifically, non-veiled women are less likely to be employed in the public sector compared to the pre-policy period. Instead, they are more likely to be employed in the private sector and as self-employed in the post-policy period. Given that the number of offered public sector jobs has remained relatively stable over time (see Figure 2), these results suggest that lifting the headscarf ban led to a shift in the composition of veiled and non-veiled women within the public sector, resulting in a higher proportion of veiled women compared to the period when veiling was not allowed.

Robustness and sensitivity check. We conduct a series of robustness and sensitivity checks to ensure the validity of our findings. First, we assess the sensitivity of our results to the definition of the post-policy period. While the headscarf ban in public institutions was officially lifted in October 2013, our data are annual, making it possible that using 2014 as the post-policy period underestimates the policy’s effect. The 2013 wave might partially reflect early responses to the policy change (see Figure 5). To address this, we re-estimate the main results excluding the 2013 wave. The coefficients, reported in Panel A of Table A7 in the Appendix, remain stable in both magnitude and precision.

Next, we evaluate concerns regarding potential selection bias in our treatment assignment based on out-of-sample predictions. Our main analysis predicts veiling status using data from the 2013 wave of the Demographic and Health Survey (DHS). While the policy change was unlikely to immediately alter veiling patterns, there is a possibility that the use of post-policy data for prediction introduces bias. To address this, we instead predict veiling status using data from the 2008 DHS wave. Results in Panel B of Table A7 confirm that our main findings remain robust, lending further credibility to our out-of-sample classification approach.

We also assess whether our estimates could be confounded by other policy changes. The gradual removal of the headscarf ban in universities began in 2010, with institutional variation in implementation. This could have influenced labor market outcomes, particularly for women born after January 1992, who were more likely to enter university after these changes. To test whether our results are driven by this cohort, we introduce a binary indicator for women born in or after January 1992 and interact it with the *Post* variable. Results in Panel C remain consistent with our main estimates, suggesting that the observed labor market effects are not confounded by earlier initiatives targeting the ban across universities.

Similarly, we control for the potential effects of the 1997 education reform, which extended compulsory schooling from five to eight years for cohorts born after 1986. This reform likely affected women’s educational attainment and subsequent labor market outcomes. To account for this, we introduce a binary indicator for women born after 1986 and interact it with the post-policy period. As reported in Panel D of Table A7, our main findings remain robust, indicating that the observed effects of the headscarf ban’s removal are distinct from those of earlier education reforms.

To further validate our results, we explore alternative treatment and control groups. In the first exercise, we redefine our treatment group as women who hold traditional attitudes toward gender roles —specifically, those who agree with the statement “*the husband should work while the wife stays home*” (i.e., traditional). If the estimated treatment effects for

women who are predicted to be veiled closely resembles that for traditional women, it would be difficult to determine whether the effect is specific to veiled women or the sample of women with traditional attitudes. In principle, given that the low correlation between being traditional towards gender role attitudes and being veiled (0.20 in the DHS sample), we should not expect similar effects of the reform on these samples. We predict traditional attitudes following the same methodology as our main analysis and re-estimate Equation (2) using the variable *Traditional* instead of *Veiled*. Results presented in Panel E indicate smaller and statistically distinct effects for traditionally-minded women. Specifically, Wald tests confirm significant differences in the estimated coefficients for employment (p-value = 0.000) and public sector employment (p-value = 0.000). These findings support that the effects of the reform are indeed specific to veiled women rather than women with traditional attitudes more generally.

In the second exercise, we use the frequency of praying daily as an additional treatment group to validate our main prediction technique. Given the strong correlation between regular prayer and veiling status (0.42 in the DHS sample), this provides a more behaviorally anchored proxy for veiling. If the estimated effects are similar for women who pray daily, this would further validate our out-of-sample predictions. The results reported in Panel F indicate significant increases in employment probability and public sector employment for women who pray regularly, along with declines in unpaid family work. Importantly, the Wald test for employment (p-value = 0.435) suggests no significant difference between effects estimated for veiled women and those who pray daily, further supporting the validity of our prediction model. However, the effect on public sector employment is larger for veiled women (p-value = 0.000), which is expected given that the reform directly addressed institutional barriers specific to veiling.

Beyond these alternative estimates, we further assess the validity of our findings using alternative statistical techniques that relax the assumption of conditional independence between veiling and labor market outcomes. Our main analysis implicitly assumes that veiling status and labor market outcomes are conditionally independent given observed covariates. However, this assumption might be violated if, for example, unobserved factors like adherence to traditional norms simultaneously influence employment decisions and veiling behavior. To relax this assumption and assess potential bias, we follow [Rubin \(1986\)](#) and [Moriarity and Scheuren \(2003\)](#) by implementing an alternative statistical matching procedure, detailed in Appendix A.2, which allows for non-zero partial covariance between veiling and employment. We perform robustness checks by assuming various levels of conditional correlation and re-estimate the difference-in-differences models. As reported in Table A8 in the Appendix, these alternative estimations yield results consistent with our main findings (Table 2). Specifically, employment and public sector employment probabilities for veiled women increased following the policy reform, while the likelihood of unpaid family work declined. The stability of these results, despite relaxing conditional independence, strengthens the robustness of our estimates.

To further validate our results, we explore alternative methods for predicting treatment status using machine learning (ML) techniques. ML approaches can capture complex, potentially nonlinear relationships between observable demographics and veiling behavior, strengthening the credibility of our identification strategy. This process involves two stages. First, we train a predictive model on the DHS dataset to establish relationships between demographic characteristics and veiling status. Second, we apply this trained model to predict veiling status for women in the HLFS dataset.

Our primary approach employs an ensemble learning method, combining predictions

from Support Vector Machines (SVM) and Random Forests (RF) through a hard-voting procedure. The trained ensemble model achieves approximately 80% accuracy in the DHS training data. Applying this model to the HLFS dataset, we find that around 63% of women are predicted to wear a headscarf, closely matching the DHS average, which further confirms the validity of our predictions. These predicted veiling statuses are then utilized in our main difference-in-differences analysis.

To test sensitivity to the choice of prediction method, we also estimate treatment effects using alternative ML models: Support Vector Machines (SVM), Random Forest (RF), K-Nearest Neighbors (K-NN), and Neural Networks (NN). Results from these models, presented in Table A12 and Table A13, are largely consistent. Across all methods, we find statistically significant increases in overall employment and public sector employment for veiled women following the policy change, coupled with declines in unpaid family work. Effects on private sector employment vary slightly across methods, with the ensemble and SVM indicating significant increases, whereas RF, K-NN, and NN produce statistically insignificant results.

Taken together, these robustness checks demonstrate that our results are not driven by arbitrary classification choices, omitted variable bias, or confounding policy changes. More importantly, our ML-based out-of-sample prediction approaches confirm that our findings hold across different methodological approaches, providing a rigorous and replicable framework for analyzing policy effects when direct treatment assignment is not observable.

4.2 Effects on Marriage Market Outcomes of Women

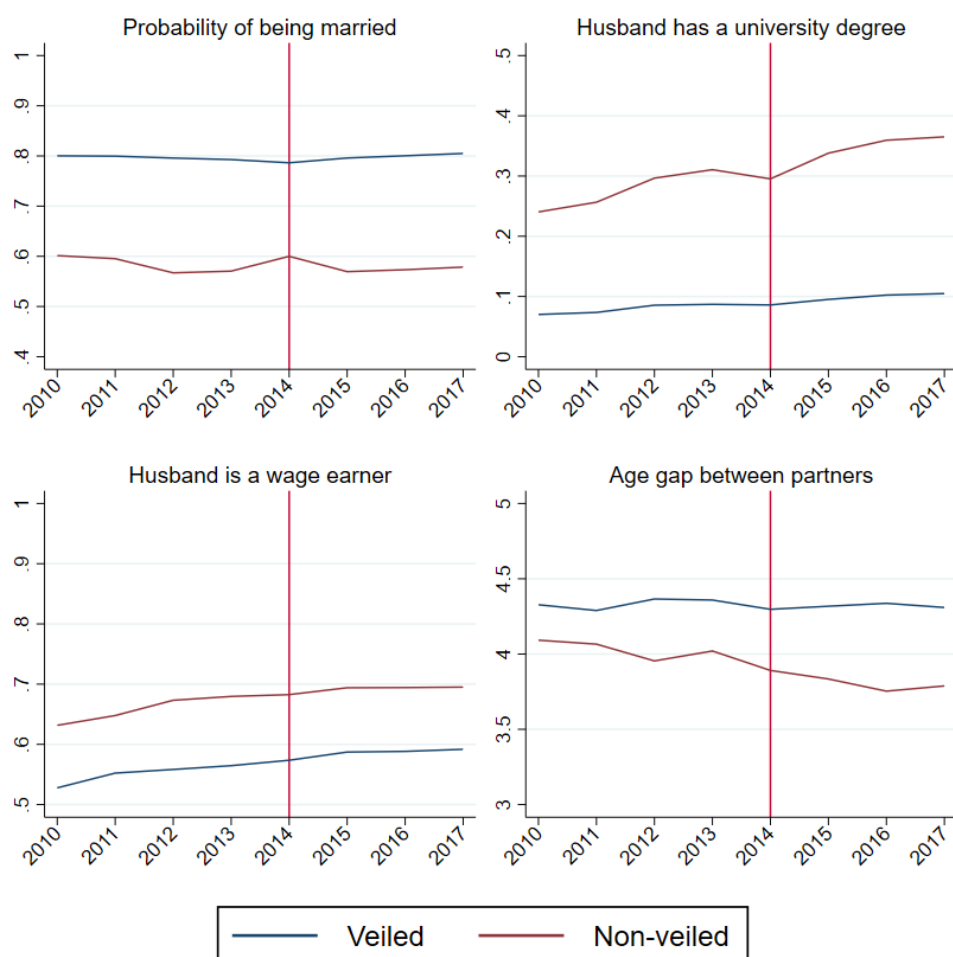
Having established the significant effects of lifting the headscarf ban on women’s labor market outcomes, we now turn to its effects on the marriage market. While labor market outcomes remain the primary focus of this paper, the reform’s potential to generate returns in the marriage market offers an additional lens to understand its broader impact beyond the labor market. For example, the removal of the headscarf ban could have altered the attractiveness of veiled women in the marriage market, potentially increasing their likelihood of marriage or improving partner quality through better sorting. Additionally, the reform may have reduced barriers to partner search by increasing veiled women’s likelihood of employment, thereby expanding their opportunities for social interaction and partner selection.

Figure 7 presents trends in key marriage market outcomes for veiled and non-veiled women from 2010 to 2017. The top-left graph illustrates trends in marriage probability, showing that the share of married women predicted to be veiled remained relatively stable at around 80%, while the marriage rate for non-veiled women slightly declined over the same period. These trends suggest differential pre-trends in marriage probabilities between the two groups. Focusing on currently married women, the remaining graphs in Figure 7 depict trends in partner quality indicators: whether the husband holds a university degree, whether the husband is a wage earner, and the age gap between spouses. The trends reveal no significant pre-policy differences in the likelihood of marrying a university-educated or wage-earning husband. However, a divergence emerges in the age gap, with the difference between veiled and non-veiled women widening after the policy change.

Table 3 provides results for the effects of the headscarf ban’s removal on marriage market outcomes. Panel A offers basic before-after estimates, while Panel B presents the the main difference-in-differences, focusing on shifts in partner characteristics and marriage patterns.

Column 1 of Panel B examines marriage probabilities for the full sample of women. The estimated effect suggests a 0.20 percentage point increase in the likelihood of marriage for veiled women following the reform. However, this effect is statistically insignificant and

Figure 7: Trends in Marriage Market Outcomes by Veiled and Non-veiled Women



Notes: Raw data from the 2010-2017 HLFS. Marriage probabilities and partner characteristics are coded as binary indicators. The age gap represents the difference in years between spouses.

small in magnitude, representing only a 0.26% increase relative to the sample mean for veiled women prior to the reform.

Focusing on currently married women, Columns 2 and 3 explore educational sorting in the marriage market. Results show no significant changes in the likelihood of veiled women marrying university-educated or high school-educated husbands, indicating stability in partner education levels. In contrast, Column 4 shows a significant 1.5 percentage point increase in the probability of veiled women marrying wage-earning husbands, reflecting shifts in partner employment characteristics. Column 5 highlights another notable finding: the age gap between veiled women and their husbands increased by 0.84 years, suggesting a preference for older partners post-reform. These results indicate that the removal of the headscarf ban influenced partner sorting, particularly by aligning veiled women with older and wage earner husbands.

In Table A9 in the Appendix, we also provide corresponding event-study estimates tracing year-by-year treatment effects. Unlike labor market outcomes, where pre-treatment trends between veiled and non-veiled women were parallel, the event study results for marriage market outcomes reveal deviations from parallel trends in the pre-reform period. Column 1 shows that veiled women were already experiencing higher marriage probabilities relative to non-veiled women in the pre-reform years, with significant positive coefficients in 2012 (0.031) and 2013 (0.022). This suggests that the observed increase in marriage probabilities for veiled women post-reform cannot solely be attributed to the policy change, as pre-existing trends were already favorable. Similarly, Columns 2 and 3 indicate no meaningful changes in

educational sorting post-reform, consistent with stable partner education levels. For partner employment characteristics, Column 4 shows that veiled women were more likely to marry wage-earning husbands compared to non-veiled women even in the pre-reform period (e.g., significant effects in 2011). However, the post-reform period intensifies this trend, with statistically significant effects observed in each year, ranging from 0.02 to 0.04. Finally, Column 5 reveals that the age gap between veiled women and their husbands was already diverging before the reform (e.g., significant effects in 2012), but this divergence further widened post-reform, with increases ranging from 0.09 to 0.18 across all post-reform years.

To address potential violations of the parallel trends assumption for certain pre-reform years, a straightforward approach is to estimate Eq.(2) by including interaction terms for the treatment dummy *Veiled* and the significant pre-reform years: 2012 and 2013 for marriage probability, 2011 and 2013 for husband’s employment as a wage earner, and 2012 for the age gap. These interactions account for deviations in pre-trends while retaining the remaining years for comparison. The results are presented in Panel A of Table A10 in the Appendix. Estimates show that the probability of marriage for veiled women, which was previously statistically insignificant, now shows a significant increase post-reform, providing some evidence that the policy may have influenced marriage probabilities. For the likelihood of husbands working as wage earners and age gap between partners, the effects remain similar to the main estimates presented in Table 3. These results confirm that the observed effects are not solely driven by differential pre-reform trends but reflect some shifts in marriage market outcomes following the policy change. However, the interpretation of these effects remains cautious, as excluding certain years limits the scope of direct comparisons across the full pre-reform period.¹⁸

While the deviations from parallel trends complicate the causal interpretation of these results, the post-reform period on average shows that veiled women were more likely to marry wage-earner men and experienced an increased age gap with their partners compared to the pre-policy period. These findings highlight the potential influence of the headscarf ban’s removal on partner selection.

To further enhance the robustness of the data matching process, we also investigate the effects on marriage market outcomes using the predicted veiling variable derived from machine learning methods, as presented in Table A13. Across all five models, we consistently observe significant increases in the age gap between spouses and in the likelihood of veiled women marrying wage-earning husbands. Importantly, nearly all models reveal no significant changes in marriage probability¹⁹ and in educational sorting, aligned with the findings reported in Table 3. The effects on husbands’ university degrees, however, are less clear-cut, with some models suggesting a marginally significant increase, while the PSM matching results does not show significant change. Given the inconsistencies across models, we conclude that the unveiling policy has minimal influence on veiled women’s spouse selection in terms of university education. This indicates that, despite some variations in model outputs, the policy’s overall effect on marriage market outcomes is clear with veiled women showing a preference for marrying older and wage-earning husbands.

All in all, our results suggest that the removal of the headscarf ban improved veiled women’s economic opportunities, particularly through increased employment and public sector jobs, enhancing their position in the marriage market. These improved labor mar-

¹⁸In addition to addressing pre-trends, we reproduced the robustness and sensitivity checks conducted for labor market outcomes (see Panels B–G in Table A10). These results are consistent with the main estimates, with one notable exception: in some specifications, the probability of marriage becomes statistically significant.

¹⁹Except for SVM; however, after applying stricter filtration in the Ensemble model, the impact on the marriage rate becomes statistically insignificant.

ket opportunities may have allowed veiled women to become more selective, favoring older and more financially stable partners—a preference aligned with traditional norms that associate older men with economic security, which is consistent with the literature (Buss, 1989; Buunk et al., 2001). However, the interpretation of these changes as causal effects of the reform is subject to potential selection effects. Veiled women marrying post-reform may differ systematically from those marrying earlier, possibly delaying marriage to focus on careers or achieving greater financial independence. Moreover, the reform may have shifted men’s preferences, increasing the desirability of veiled women as partners and influencing marriage market outcomes. While these factors affect how we interpret the impact on the marriage market, the evidence points to notable shifts in marriage patterns among veiled women following the policy change.

Table 3: *Effects of the Headscarf Ban Removal on Women’s Marriage Market Outcomes*

Outcome	Marriage probability	Husband has a university degree	Husband has a high school degree	Husband is a wage earner	Age gap between partners
	(1)	(2)	(3)	(4)	(5)
Panel A: Before-After Estimates					
Post	0.011*** (0.001)	0.006*** (0.001)	0.004*** (0.001)	0.013*** (0.001)	0.039*** (0.012)
Year Fixed Effects	No	No	No	No	No
Observations	934,060	648,747	648,747	648,747	648,747
Mean Dependent Var.	0.70	0.14	0.21	0.84	4.23
Panel B: Difference-in-Differences Estimates					
Veiled $\times$ Post	0.002 (0.002)	0.002 (0.002)	-0.001 (0.003)	0.015*** (0.002)	0.084*** (0.026)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	934,060	648,747	648,747	648,747	648,747
Mean Dependent Var.	0.78	0.08	0.20	0.81	4.24

Notes: Data are from the 2010-2017 HLFS. Column (1) uses the full sample of women, while Columns (2)–(5) are restricted to currently married women. The last row in each panel reports the mean outcome for veiled women before the policy change. Regressions control for nativity, education (categorical), age, age squared, household size, and region and year fixed effects, along with their interactions. Standard errors (in parentheses) are clustered at the individual level. All estimates are weighted using cross-sectional survey weights from the wave in which the outcome was measured. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

At last, the observed shifts in labor and marriage market outcomes raise important questions about their implications for household decision-making, particularly the gendered division of market work. While the exact mechanisms remain unclear—since the reform affected partner characteristics—we examine its impact on the gender labor supply gap within households.

Panel B of Table A11 presents difference-in-differences estimates for married women, analyzing changes in working hours and earnings. Columns 1–3 show that veiled women increased their weekly working hours by 0.63 hours more than their non-veiled counterparts post-reform, while their husbands’ working hours rose by 0.39 hours. However, as Columns 4–5 indicate, hourly earnings for both spouses remained unchanged, suggesting that the increase in labor supply was not driven by an income effect. Despite these shifts, the gen-

der labor supply gap within households did not significantly narrow, implying that veiled women’s employment gains were not large enough to alter the broader division of market work between spouses.

These changes in labor supply patterns may reflect shifts in intra-household bargaining power. Economic theory suggests that factors improving women’s relative standing—such as greater access to employment—enhance their decision-making power within households (Chiappori et al., 2002). Existing evidence shows that labor market opportunities of women increase their bargaining power within the household (Friedberg and Webb, 2006; Majlesi, 2016). In this context, the unveiling reform likely strengthened veiled women’s bargaining position by expanding their employment opportunities.

If veiled women gained bargaining power, they might have influenced household decisions in ways that affected both their own and their husbands’ labor supply. However, without data on assignable goods or intra-household spending, we cannot directly measure changes in decision-making power. This remains an open question for future research.

4.3 Heterogeneous Effects

In this section, we examine whether the observed effects of lifting the headscarf ban vary by women’s education level, marital status, and regional disparities in the prevalence of veiling. Exploring these heterogeneous effects sheds light on how different socioeconomic and demographic groups responded to the policy change, providing a nuanced understanding of its implications.

We begin by assessing whether the effects of the policy change on labor market outcomes differ by women’s educational attainment. To this end, we include a dummy variable, *HigherDegree*, which equals 1 if the woman has completed at least a high school education or higher, and 0 otherwise. This variable is interacted with the difference-in-differences term, $Veiled \times Post$, to estimate heterogeneous effects based on education level. Results in Panel A of Table 4 indicate no significant heterogeneity by education level across any labor market outcome. The estimated coefficients are close to zero and statistically insignificant, suggesting that the removal of the headscarf ban expanded labor market opportunities for veiled women across all educational backgrounds. This finding is particularly relevant for public sector employment, where we observe no evidence that hiring favored more educated veiled women. Instead, the results suggest that the reform eliminated institutional barriers rather than selectively benefiting women with higher qualifications.

Next, we examine whether the policy effects differ by marital status. We introduce a binary variable, *Married*, which equals 1 if the woman is currently married and 0 otherwise, and interact it with $Veiled \times Post$. The estimated coefficients in Panel B are small and statistically insignificant for most labor market outcomes, indicating that the overall effects of the reform did not vary substantially between married and unmarried veiled women. However, married veiled women became significantly less likely to work in the private sector following the reform, relative to their unmarried counterparts. This suggests that a shift from private to public sector jobs among employed married veiled women, possibly due to the greater job security and benefits associated with public sector employment.

We then explore whether the labor market effects of lifting the headscarf ban differ based on the regional prevalence of veiled women. Using data from the 2013 DHS, we construct a continuous measure representing the proportion of veiled women in each NUTS2 region and define a binary variable, *HighShare*, which equals 1 if the region’s veiling prevalence is above the median (0.60) and 0 otherwise.

Results in Panel C of Table 4 indicate no significant differences in employment probabilities or public sector employment between women residing in high- and low-veiling regions. Similarly, private sector employment does not exhibit significant variation across these regions. However, we observe a small but significant decrease in self-employment in high-veiling regions, suggesting that in areas where veiled women are more socially accepted, the reform reduced the necessity of self-employment—which may have previously served as an alternative to bypass workplace restrictions. Additionally, we find a statistically significant positive coefficient on unpaid family work, indicating that the overall decline in unpaid family work was primarily driven by women in low-veiling regions. This suggests that veiled women in regions where they were a minority were more likely to transition out of unpaid family roles and into formal employment following the reform.

These findings have broader implications for contexts where veiled women are under-represented, such as Western European countries with veiling bans. If the effects of lifting the ban do not vary substantially based on veiling prevalence within Turkey, this suggests that removing institutional barriers could have significant effects even in settings where veiled women face greater social exclusion. While we cannot directly extrapolate our findings to the European context, they highlight the potential for policy changes to shape labor market outcomes in contexts where veiling intersects with broader social and institutional constraints.

Finally, one might wonder whether the observed labor market effects of lifting the ban were primarily driven by regions where Islamist political parties hold more influence, rather than reflecting a broader labor market shift. If municipalities with Islamist mayors were more inclined to hire veiled women after the reform—either due to ideological preferences or shifts in local hiring policies—this could raise concerns that our findings partially reflect political favoritism rather than the removal of institutional barriers.

Prior research by Corekcioglu (2021) examines this question in the context of municipal jobs, showing that the repeal of the ban increased female employment only in municipalities where Islamist parties narrowly won elections. To explore this possibility, we leverage variation in Islamist political influence across regions by constructing a measure based on the vote share of Islamic political parties (Justice and Development Party and Felicity Party) in the 2014 municipal elections. We classify regions into high and low Islamic vote-share areas using the median regional vote share (46.8) and interact this measure with the difference-in-differences term to test whether the labor market effects of the policy change are concentrated in regions where Islamist parties hold stronger electoral support.

Results in Panel D of Table 4 show that the main effect of the reform on veiled women’s employment remains statistically significant, confirming that the overall labor market shifts are not exclusively driven by Islamist mayor controlled municipalities. However, we find that veiled women in high Islamic vote-share regions experienced larger employment gains, particularly in public sector jobs. This suggests that Islamist-leaning municipalities may have played a role in facilitating access to public sector employment following the policy change. Notably, we find no significant differences in private sector employment, indicating that the private labor market effects were not shaped by regional political affiliations.

Taken together, these findings suggest that while Islamist-controlled regions may have amplified the employment effects, the overall increase in labor market participation among veiled women extends beyond these regions. This aligns with our broader argument that lifting the headscarf ban removed institutional barriers rather than merely redistributing job opportunities based on political alignment. While local political dynamics may have influenced the extent of public sector hiring, the evidence indicates that the policy had

Table 4: *Heterogeneous Effects of the Headscarf Ban Removal on Women’s Labor Market Outcomes*

Outcome	Employment probability (1)	Public sector employment (2)	Private sector employment (3)	Self-employed (4)	Unpaid family worker (5)
Panel A: Education level					
Veiled $\times$ Post $\times$ HigherDegree	0.004 (0.005)	0.004 (0.003)	0.001 (0.005)	0.001 (0.002)	0.001 (0.002)
Panel B: Being married					
Veiled $\times$ Post $\times$ Married	-0.001 (0.005)	0.003 (0.002)	-0.009** (0.004)	0.001 (0.002)	-0.001 (0.002)
Panel C: Regional share of veiled women					
Veiled $\times$ Post $\times$ HighShare	0.006 (0.004)	-0.002 (0.002)	-0.001 (0.004)	-0.003* (0.002)	0.006** (0.003)
Panel D: Political alignment (Islamic party vote share)					
Veiled $\times$ Post	0.005* (0.003)	0.012*** (0.001)	0.002 (0.003)	-0.002 (0.001)	-0.012*** (0.002)
Veiled $\times$ Post $\times$ HighVoteShare	0.013*** (0.005)	0.006*** (0.002)	-0.001 (0.004)	0.002 (0.002)	0.003 (0.002)
HighVoteShare	0.115*** (0.007)	-0.060*** (0.003)	0.154*** (0.005)	0.003 (0.003)	0.006 (0.004)
Observations	934,060	934,060	934,060	934,060	934,060

Notes: Data are from the 2010-2017 HLFS. Regressions control for whether the woman is a native, education dummies (except in Panel A), age, age squared, household size, and region and year fixed effects, with their interactions. Standard errors are clustered at the individual level. Data are weighted using cross-sectional weights from the wave in which the outcome was measured. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

widespread labor market effects beyond politically favorable municipalities.

Table 5 presents heterogeneous effects for marriage market outcomes. Results in Panel A indicate that more educated veiled women—those with at least a high school degree—were significantly less likely to marry a university educated husband, with an estimated 1.9 percentage point decline, while they experienced a 1.1 percentage point increase in the likelihood of marrying a husband with a high school degree. Given that university attainment among veiled women is extremely low, suggesting a pattern of educational homogamy. Moreover, the positive and significant effect on the age gap indicates that the increase in the age gap between spouses was more pronounced for more educated veiled women (+0.13 years). Given that more educated women are typically expected to be less traditional, this finding is somewhat unexpected. One possible explanation is that traditional partner preferences, such as marrying an older spouse, remain relevant even among more educated veiled women.

Results in Panel B suggest that marriage probabilities increased more in high-veiling regions, where veiled women were 1.1 percentage points more likely to marry after the reform compared to those in low-veiling regions. However, there are no significant differences in partner education, employment status, and age gap, indicating that the key shifts in partner characteristics were not region-specific. This pattern mirrors our labor market findings, where the employment effects of the reform were consistent across regions, regardless of veiling prevalence.

Finally, results in Panel C indicate that the overall marriage market effects remain robust after accounting for regional political alignment. The coefficients on *Veiled $\times$ Post* remain similar to those in Table 4, confirming that the reform’s effects on partner selection were not primarily driven by Islamist-leaning municipalities. While veiled women in high Islamic vote-share regions were slightly more likely to marry, we find no significant differences in partner characteristics, showing that the observed shifts in marriage market outcomes were not politically driven.

Table 5: *Heterogeneous Effects of the Headscarf Ban Removal on Marriage Market Outcomes*

Outcome	Marriage probability (1)	Husband has a university degree (2)	Husband has a high school degree (3)	Husband is a wage earner (4)	Age gap (5)
Panel A: Education level					
Veiled $\times$ Post $\times$ HigherDegree	-0.001 (0.004)	-0.019*** (0.006)	0.011* (0.006)	-0.006 (0.006)	0.134** (0.057)
Panel B: Regional share of veiled women					
Veiled $\times$ Post $\times$ HighShare	0.011** (0.004)	0.002 (0.004)	0.006 (0.005)	-0.010 (0.006)	-0.047 (0.056)
Panel D: Political alignment (Islamic party vote share)					
Veiled $\times$ Post	-0.003 (0.002)	-0.000 (0.002)	-0.001 (0.003)	0.019*** (0.004)	0.099*** (0.034)
Veiled $\times$ Post $\times$ HighVoteShare	0.011*** (0.003)	0.005 (0.004)	-0.001 (0.005)	0.009 (0.006)	-0.031 (0.053)
HighVoteShare	0.025*** (0.008)	-0.065*** (0.007)	-0.078*** (0.011)	0.093*** (0.013)	-0.448*** (0.150)
Observations	934,060	648,747	648,747	648,747	648,747

Notes: Data are from the 2010-2017 HLFS. While the analysis in column 1 includes all women, the rest includes the sample of currently married women. Regressions control for whether the woman is a native, education dummies (except in Panel A), age, age squared, household size, and region and year fixed effects, with their interactions. Standard errors are clustered at the individual level. Data are weighted using cross-sectional weights from the wave in which the outcome was measured. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

5 Conclusion

This paper examines the impact of lifting Turkey’s headscarf ban on women’s labor and marriage market outcomes. Using two complementary datasets—the Turkish Household Labor Force Survey (HLFS) and the Turkish Demographic and Health Survey (DHS)—we develop novel out-of-sample predictions to identify veiled women, leveraging demographic characteristics via statistical matching and machine learning methods. We then use a difference-in-differences strategy to estimate the causal effects of removing the ban.

Our findings show significant positive effects of the reform on labor market outcomes for veiled women. Specifically, lifting the ban increased veiled women’s probability of employment, predominantly driven by greater access to public-sector jobs. The reform significantly raised the probability of public sector employment, suggesting that institutional discrimination against veiled women was reduced. We also document a statistically significant decrease in unpaid family work, indicating a shift toward formal employment with better job security and economic stability. Moreover, the reform increased representation of veiled women in occupations like technician and associate professional roles, particularly benefiting women with lower levels of education.

We also uncover meaningful effects on marriage market outcomes. Following the reform, veiled women experience shifts in partner sorting: they marry partners who tend to be older and are more likely to be employed. This highlights that lifting the ban not only affects individual employment decisions but also alters marriage patterns and partner selection dynamics.

Importantly, we find that the observed labor and marriage market effects are similar across regions, regardless of local variations in veiling prevalence. These results underline that removing institutional barriers to veiling can yield substantial social and economic benefits, even in contexts where social exclusion of veiled women is prevalent. Given ongoing debates around headscarf regulations in European and other countries, our findings offer valuable policy implications regarding how similar reforms could enhance labor force partic-

ipation, economic integration, and social inclusion of religious minority women elsewhere.

Our study points toward several avenues for future research. First, examining long-term impacts on veiling and educational decisions among young women would deepen understanding of how institutional reforms shape cultural and religious behaviors. Second, investigating changes in intra-household bargaining power and family decision-making following such policy reforms could shed light on broader implications for gender equality within families. Third, assessing the productivity impacts on public sector employing newly integrated veiled women would offer further insights into organizational outcomes associated with increased diversity and inclusion.

Overall, our findings provide robust evidence that removing legal restrictions on veiling significantly improves labor market integration, reduces gender disparities, and influences marriage dynamics positively. Turkey's experience thus serves as an informative reference point for other countries considering or debating policies around religious attire, providing clear evidence on the socio-economic implications of lifting discriminatory institutional barriers.

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A Appendix

A.1 Machine learning techniques

Recent studies have increasingly focused on applying machine learning (ML) techniques to address missing value issues (Chen and McCoy, 2024) and improve out-of-sample predictions (Athey and Imbens, 2019). Compared to conventional statistical methods, ML approaches offer advantages such as flexible functional forms, greater time efficiency and higher accuracy (Mullainathan and Spiess, 2017). Given the common challenges of poor data quality in various fields, multiple ML methods have been proposed to tackle missing value problems in diverse settings (Gogas and Papadimitriou, 2021). Applications in economics include examples from healthcare data (Mullainathan and Obermeyer, 2022), asset pricing (Goldstein et al., 2021), and other areas.

Machine learning approaches are typically categorized into supervised, unsupervised, and semi-supervised learning. The key distinction lies in whether labeled datasets are available—that is, whether clear relationships between input features and corresponding output are established. In this paper, we use supervised learning, as we have observed demographic characteristics as inputs and veiling status as outputs derived from the DHS dataset. Supervised learning achieves high accuracy when the training dataset is abundant and well-labeled. Since the DHS is a high-quality, nationally representative, and randomized dataset, we can reasonably expect strong performance from the applied ML techniques.

The ML procedure is divided into two main steps: training and prediction. In the training phase, we analyze the DHS dataset to identify patterns between observed demographic characteristics and the headscarf-wearing behavior of female respondents. Using a selected ML algorithm, we develop a model capable of predicting headscarf-wearing outcomes for unseen inputs. To train the model, we split the DHS dataset into training and test sets. The algorithm uses the training set to fit the model, and its predictions are compared against the true outputs in the test set. The model’s parameters are adjusted iteratively to minimize the difference between its predictions and the actual outcomes. In the prediction phase, the optimized model is applied to the HLFS dataset to predict unknown veiling outcomes based on observed demographic inputs. Several key algorithms have demonstrated strong performance in recent studies. ²⁰

A.1.1 Ensemble Learning

Ensemble learning integrates multiple models to enhance accuracy, robustness, and flexibility by combining their predictions (Breiman, 1996). It is particularly useful when different algorithms capture distinct data patterns or when a single model’s accuracy is insufficient, helping to mitigate overfitting in small or noisy datasets.

A variety of models can be used in ensemble learning, with the most common being Support Vector Machines, Random Forests, Linear Models, Neural Networks, and others.

For our robustness check in dataset matching using machine learning techniques, we employ an ensemble approach combining Support Vector Machines and Random Forests as our primary model. The following sections provide a detailed explanation of these models.

²⁰In the training and prediction process, the machine learning models do not discard samples. As a result, the number of observations and mean dependent values reported in the result tables using ML-imputed data may differ slightly from those using PSM. However, this should not affect the sample composition.

A.1.2 Support Vector Machines

Support Vector Machines (SVM) is a powerful algorithm for data classification ([Cortes, 1995](#)). Its core principle is to find an optimal decision boundary, known as a hyperplane, that separates two classes of data points. The objective is to maximize the margin—the distance between the hyperplane and the nearest data points from each class, called support vectors—while ensuring accurate classification.²¹

Consider a two-class classification problem with training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where $\mathbf{x}_i$ are feature vectors and $y_i \in \{-1, 1\}$ are class label.

In the primal form of SVM, the goal is to find a hyperplane that optimally separates the two classes. The decision function is a linear function defined as:

$$f(x) = \mathbf{w} \cdot \mathbf{x} + b$$

where $\mathbf{w} \in \mathbb{R}^n$ is the weight vector (orthogonal to the decision hyperplane), $\mathbf{x} \in \mathbb{R}^n$ is the input feature vector, and $b \in \mathbb{R}$ is the bias term. The dot product $\mathbf{w} \cdot \mathbf{x}$ represents the projection of the data point x onto vector w , which plays a key role in defining the decision boundary.

For classification, the predicted class is determined by the sign of $f(x)$. If $f(x) > 0$, predict class is +1; and if $f(x) < 0$, predict class is -1.

The primal optimization problem aims to find the weight vector w and bias term b that define the optimal separating hyperplane. This is formulated as:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2$$

subject to the classification constraints:

$$y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 \quad \text{for all } i$$

The objective function $\frac{1}{2} \|\mathbf{w}\|^2$ is minimized to maximize the margin between the two classes, while the constraints ensures that each data point is classified correctly.

When the data is non-linearly separable, the approach is to map the data into a higher-dimensional space using a feature mapping function $\phi(x)$. In this higher-dimensional space, the data becomes linearly separable, and the decision function is given by:

$$f(x) = \mathbf{w} \cdot \phi(\mathbf{x}) + b$$

Instead of explicitly computing $\phi(x)$, the kernel trick is used to replace the dot product $\mathbf{w} \cdot \mathbf{x}_i$ with a kernel function $K(\mathbf{x}_i, \mathbf{x}_j)$, which directly computes the similarity between data points in the higher-dimensional space.

The dual form of the SVM optimization problem is then written in terms of the Lagrange multipliers α_i :

$$\max_{\alpha_i} \left(\sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i,j=1}^N \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \right)$$

subject to the constraints:

$$\sum_{i=1}^N \alpha_i y_i = 0 \quad \text{and} \quad 0 \leq \alpha_i \leq C$$

²¹Maximizing the margin helps create a more robust decision boundary, making it less sensitive to noise and outliers.

Here the C is a regularization parameter that controls the trade-off between maximizing the margin and minimizing classification errors. $K(\mathbf{x}_i, \mathbf{x}_j)$ is the kernel function. We have several commonly used kernel functions include: linear, polynomial and radial basis function Kernel.

As shown, a key advantage of SVM is its flexibility in handling complex data structures by employing different kernel functions, making it well-suited for nonlinear classification problems. After performing a grid search to optimize model parameters for maximum training accuracy, we configure the SVM model as follows:

```
svm_model = SVC(kernel='rbf', C=10, gamma='scale')
```

Here, we use the Radial Basis Function (RBF) kernel to capture nonlinear decision boundaries. The gamma parameter is set to 'scale', adapting to the data to balance complexity and prevent overfitting or underfitting. Additionally, the regularization parameter $C = 10$, emphasizing classification accuracy over margin maximization, allowing the model to handle misclassified points with lower tolerance.

A.1.3 Random Forest

Random Forest (RF) is a powerful machine learning algorithm known for its ability to identify important features and its robustness to noise and outliers, making it highly resistant to overfitting (Breiman, 2001). It is a reliable choice for a variety of tasks, especially classification. The implementation involves several key steps:

The first step is “Bootstrap Sampling”. We generate multiple bootstrap samples from the original dataset. Each sample $D_{\text{bootstrap}}$ is a random subset created by sampling with replacement. ²²

$$D_{\text{bootstrap}} = \{\text{Sample from } D \text{ with replacement}\}$$

where D is the original dataset.

In the second step “Training Decision Trees”, a decision tree is independently trained on each bootstrap sample $D_{\text{bootstrap}}$. Here’s how the training process works:

At each node of the tree, the algorithm evaluates which feature and corresponding split threshold best separates the data into subsets. This is done by assessing potential splits using impurity measures such as Gini impurity or entropy. The goal is to find the feature and threshold that minimize the impurity, creating more homogeneous subsets.

$$Gini(D) = 1 - \sum_{i=1}^C p_i^2$$

where p_i is the proportion of class i in the node, and C is the number of classes. A lower Gini impurity means the node is more pure (i.e., most of the samples in the node belong to the same class), while a higher Gini impurity indicates that the samples are more mixed across different classes.

Once the best feature and threshold are identified for the current node, the dataset is split into two child nodes based on the chosen feature and value. This process is repeated recursively for each child node in the same way, each time selecting the best feature to further split the data, until a stopping criterion is met. Some common stopping criteria include: maximum depth of the tree, minimum samples per leaf, minimum impurity split and etc.

The third step is “Prediction”. Once all trees are trained, predictions are made by aggregating the outputs of all trees. A “majority vote” is taken to determine the final class

²²Some data points may be repeated, and some may be left out.

label. The class that receives the most votes from the individual trees is selected as the final prediction:

$$\hat{y} = \text{mode}(\hat{y}_1, \hat{y}_2, \dots, \hat{y}_T)$$

where $\hat{y}_i$ is the predicted class from tree i , and T is the total number of trees.

After performing a grid search to find the optimal parameters for the best accuracy, we configure the RF model as follows:

- `n_estimators = 100`: This specifies the number of trees in the random forest. A higher number improves the model's robustness but also increases computation time.

The following three parameters are stopping criteria that control how the individual decision trees are trained:

- `max_depth = 10`: This sets the maximum depth of each tree. Limiting depth prevents the trees from becoming too complex and overfitting to the training data.

- `min_samples_split = 5`: This defines the minimum number of samples required to split an internal node. Nodes with fewer than 5 samples cannot be split further. This helps in reducing overfitting by ensuring that splits are based on sufficiently large subsets of data.

- `min_samples_leaf = 4`: This controls the minimum number of samples required in each leaf node after a split. It ensures each leaf contains at least 4 samples, which prevents overly specific splits and helps avoid overfitting.

```
rf_model = RandomForestClassifier(
    n_estimators=100,
    max_depth=10,
    min_samples_leaf=4,
    min_samples_split=5
)
```

A.1.4 Other ML models

As part of our sensitivity analysis, we implement two other commonly used ML models and present the diff-in-diff results using imputed data from these models.

K-Nearest Neighbor (K-NN) is a simple yet powerful instance-based algorithm ([Aha et al., 1991](#)). Unlike other models, K-NN doesn't require an explicit training phase. Instead, it stores the training dataset and uses it directly for prediction. To predict the output for a new data point, K-NN calculates the distance between the new point and all existing points in the training set, typically using Euclidean distance or another distance measure. It then identifies the K-nearest neighbors and classifies or regresses based on the labels or values of these neighbors. We select the optimal parameters for the model as follows:

```
knn_model = KNeighborsClassifier(
    n_neighbors=15,
    algorithm = 'auto',
    leaf_size=30,
    metric = 'minkowski',
    n_jobs=-1,
    p=1,
    weights = 'uniform'
)
```

Neural Networks (NN) are powerful machine learning models inspired by the structure and functioning of the human brain. They consist of interconnected nodes, or neurons,

organized into layers, which allow the model to recognize patterns and learn complex relationships in data. Neural networks are designed to learn from data through a series of connected layers, with each layer performing mathematical operations to transform input data into useful outputs (Rumelhart et al., 1986).

For our sensitivity analysis, we establish a neural network model with the following architecture:

- **Input Layer:** Contains $X_{train.shape}[1]$ neurons, equal to the number of input features.
- **Hidden Layers:** Two hidden layers with 64 and 32 neurons. These layers use the ReLU activation functions to introduce non-linearity.
- **Output Layer:** one neuron with a sigmoid activation function to produce binary classification probabilities.

The model is compiled using the Adam optimizer and binary cross-entropy loss, and we monitor accuracy during training to assess performance.

```
model = Sequential()
model.add(Dense(
    64, input_dim=X_train.shape[1], activation='relu'))
model.add(
    Dense(32, activation='relu'))
model.add(
    Dense(1, activation='sigmoid'))

model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy'])
```

A.2 Statistical Prediction Methods

Statistical prediction is a valuable tool for matching datasets with missing information. This method helps explore relationships between variables from different samples and enables the creation of a single dataset with predicted variables from statistical models.

As a robustness check to our results using imputation by PSM, we apply the statistical matching technique proposed by Rubin (1986) and later refined by Moriarity and Scheuren (2003). This method is particularly useful for addressing the conditional independence assumption. Most imputation techniques implicitly assume that veiling behaviors and labor market outcomes are conditionally independent given commonly observed variables. However, assuming that headscarf-wearing behaviors and labor market outcomes are independent given the same set of characteristics seems strong. This assumption, if incorrect, could lead to biased estimates. The alternative statistical matching method originally proposed by Rubin addresses this issue by generating predicted veiling info based on an assumed partial correlation between veiling and labor market outcomes. This partial correlation can be informed by prior literature or other datasets that jointly observe both variables.

There are two main steps in Rubin’s approach. In the first step, we estimate the predicted missing values of veiling. We separately regress the dummy variable for veiling (Y) and labor

participation (X) on observed characteristics Z using the following equations:

$$\begin{aligned} Y &= a_0 + aZ + \epsilon \\ X &= b_0 + bZ + \mu \end{aligned} \tag{3}$$

In our practice, the commonly observed characteristics Z include a woman's age, education levels, a dummy indicating whether the woman is native, region dummies at the NUTS2 level, and household size.

Using the regression results, we construct the following covariance matrix:

$$\begin{pmatrix} \mathbf{0} & \boldsymbol{\alpha} & \boldsymbol{\beta} \\ -\boldsymbol{\alpha}^T & pvar_{Y|Z} & \sigma_{Y,X|Z} \\ -\boldsymbol{\beta}^T & \sigma_{Y,X|Z} & pvar_{X|Z} \end{pmatrix} \tag{4}$$

where $\boldsymbol{\alpha} = [a, 0]^T$ and $\boldsymbol{\beta} = [b, b_0]^T$ are estimated coefficient vectors. The terms $-\boldsymbol{\alpha}^T$ and $-\boldsymbol{\beta}^T$ represent the corresponding negative coefficients row vectors. $pvar_{Y|Z}$ and $pvar_{X|Z}$ denote the partial variances of (Y, Z) and (X, Z) , respectively, which are computed from the variance of the residuals in the two regressions. The partial covariance $\sigma_{Y,X|Z} = \rho_{Y,Z|X}$ is derived from the DHS dataset wave 2013, which includes both veiling info and labor supply-related variables. This partial covariance helps capture the conditional relationship between veiling and labor participation while accounting for the observed characteristics Z .

We then apply the sweep matrix 4 to the two dependent variables in regression equations 3. Through this process, we obtain the regression coefficients of Y on $(1, X, Z)$ and those of X on $(1, Y, Z)$. These two coefficients are then used to predict the missing veiling variable in both DHS and HLFS. Since the partial correlation $\rho_{Y,Z|X}$ is assumed to be positive, the predicted values incorporate the conditional dependence between Y and Z .²³

In the second step, we match the two samples. Specifically, each respondents in HLFS who lacks veiling info is assigned a predicted value $\hat{Y}$, and is then matched with the respondent in DHS who has the closest predicted value, conditional on identical characteristics.

²³The prediction is implemented using STATA commands proposed by [Alpman \(2016\)](#).

A.3 Tables

Table A1: Occupations in the Public Sector (%)

	Men	Women
Legislators and Senior Officials	4.06	3.02
Professionals	41.09	57.81
Technicians and Associate Professionals	12.49	12.71
Clerks	14.69	17.34
Service Workers and Shop and Market Sales Workers	15.14	6.03
Skilled Agricultural and Fishery Workers	0.29	0.11
Craft and Related Trades Workers	2.59	0.37
Plant and Machine Operators and Assemblers	2.44	0.08
Elementary Occupations	7.19	2.54
Total	100.00	100.00

Notes: The share of men and women in each occupation is calculated using employed samples from the 1990 British Labor Force Survey. We follow the SIC-92 classifications.

Table A2: *Summary Statistics Using The HLFS Dataset*

	All sample	Non-veiled	Veiled
	Mean (S.D) (1)	Mean (S.D) (2)	Mean (S.D) (3)
Panel A: Background characteristics			
Age	32.88 (8.906)	31.46 (8.564)	33.79 (9.000)
Native	0.978 (0.148)	0.960 (0.196)	0.989 (0.105)
University degree	0.151 (0.358)	0.322 (0.467)	0.042 (0.200)
High school degree and above	0.352 (0.478)	0.640 (0.480)	0.169 (0.374)
Primary school degree and below	0.494 (0.499)	0.228 (0.420)	0.664 (0.472)
Married	0.713 (0.452)	0.582 (0.493)	0.797 (0.402)
Household size	3.925 (1.914)	3.235 (1.310)	4.271 (2.070)
Panel B: Labor market outcomes			
Employed	0.330 (0.470)	0.395 (0.489)	0.289 (0.453)
Employed as wage earner	0.222 (0.415)	0.339 (0.474)	0.147 (0.354)
Public sector employment	0.051 (0.221)	0.102 (0.302)	0.019 (0.138)
Self-employed	0.034 (0.184)	0.0325 (0.177)	0.036 (0.187)
Unpaid family worker	0.083 (0.276)	0.044 (0.204)	0.108 (0.311)
Obs.	935,476	341,378	592,682
Panel C: Partner and household characteristics			
<i>Conditional on being married:</i>			
Partner's age	39.55 (8.572)	38.74 (8.163)	39.96 (8.742)
Age gap between partners	4.191 (4.380)	3.922 (4.173)	4.326 (4.475)
Partner has a university degree	0.162 (0.368)	0.308 (0.462)	0.088 (0.284)
Partner has a high school degree	0.379 (0.485)	0.564 (0.496)	0.286 (0.452)
Partner is employed	0.831 (0.375)	0.858 (0.349)	0.817 (0.387)
Partner is a wage earner	0.604 (0.489)	0.675 (0.468)	0.568 (0.495)
Obs.	650,163	203,785	444,962

Notes: Data are from the Turkish HLFS 2010-17. The sample covers all married women aged 25-50. The table presents the means, and standard deviations (in parenthesis) of selected variables for all samples in column 1, for non-veiled women in column 2, and for veiled women in column 3, respectively. Data are weighted using the cross-sectional weights for the wave at which the outcome was measured.

Table A3: *Summary Statistics For Veiled and Non-veiled Women Using The 2013 DHS*

	Non-veiled	Veiled	
	(1)	(2)	(3)
	Mean	Mean	Difference
	(S.D)	(S.D)	(1)-(2)
Panel A: Background characteristics			
Age	31.17 (8.631)	33.61 (8.799)	-2.446*** (0.234)
Native	0.951 (0.215)	0.989 (0.106)	-0.037*** (0.005)
University degree	0.387 (0.487)	0.067 (0.250)	0.320*** (0.011)
High school degree and above	0.468 (0.499)	0.109 (0.311)	0.359*** (0.012)
Primary school degree and below	0.204 (0.008)	0.645 (0.007)	-0.441*** (0.011)
Married	0.616 (0.487)	0.841 (0.366)	-0.225*** (0.012)
Panel B: Labor market outcomes			
Employed	0.427 (0.495)	0.268 (0.443)	0.159*** (0.013)
Employed as wage earner	0.334 (0.472)	0.0839 (0.277)	0.250*** (0.011)
Employed in public sector	0.095 (0.293)	0.012 (0.108)	0.083*** (0.006)
Self-employed	0.038 (0.190)	0.054 (0.227)	-0.016*** (0.005)
Unpaid family worker	0.022 (0.147)	0.091 (0.287)	-0.068*** (0.005)
Obs.	2,793	6,005	
Panel C: Partner and household characteristics			
<i>Conditional on being married:</i>			
Partner's age	38.64 (8.193)	39.18 (8.891)	-0.542** (0.270)
Age gap between partners	4.049 (4.568)	4.373 (4.429)	-0.324** (0.148)
Partner has a university degree	0.318 (0.466)	0.010 (0.299)	0.218*** (0.014)
Partner has a high school degree	0.576 (0.494)	0.288 (0.453)	0.288*** (0.015)
Partner is employed	0.931 (0.253)	0.907 (0.291)	0.024*** (0.008)
Partner is a wage earner	0.529 (0.499)	0.470 (0.499)	0.059*** (0.016)
Household size	3.752 (1.335)	5.067 (2.296)	-1.314*** (0.047)
Obs.	1,729	5,071	

Notes: The data are from the 2013 Turkish Demographic and Health Survey. Sample covers all women aged 18-49. The table presents the means, and standard deviations (in parenthesis) of selected variables for non-veiled women in column 1, and for veiled women in column 2, and the difference between non-veiled and veiled women in column 3. Descriptive statistics in Panel C are conditional on being samples currently being married. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

Table A4: *Event Study Estimates for The Effects of The Policy Change on The Labor Market Outcomes of Women*

Outcome	Employment probability	Public sector em- ployment	Private sector em- ployment	Self- employed	Unpaid family worker
	(1)	(2)	(3)	(4)	(5)
Event Study Estimates					
Veiled $\times$ 2011	0.005 (0.004)	0.001 (0.002)	-0.000 (0.003)	0.001 (0.002)	0.005** (0.002)
Veiled $\times$ 2012	-0.007 (0.004)	0.000 (0.002)	-0.006* (0.004)	0.005*** (0.002)	0.001 (0.002)
Veiled $\times$ 2013	-0.004 (0.004)	0.006*** (0.002)	-0.006 (0.004)	0.003* (0.002)	-0.004* (0.002)
Veiled $\times$ 2014	0.012*** (0.005)	0.011*** (0.002)	0.005 (0.004)	-0.001 (0.002)	-0.005** (0.002)
Veiled $\times$ 2015	0.004 (0.005)	0.020*** (0.002)	-0.010*** (0.004)	0.003 (0.002)	-0.012*** (0.002)
Veiled $\times$ 2016	0.011** (0.005)	0.018*** (0.002)	-0.000 (0.004)	0.004** (0.002)	-0.014*** (0.002)
Veiled $\times$ 2017	0.009** (0.005)	0.019*** (0.002)	-0.002 (0.004)	0.002 (0.002)	-0.012*** (0.002)
Observations	934,060	934,060	934,060	934,060	934,060
Mean dependent var.	0.28	0.02	0.11	0.04	0.11

Notes: Data are from the HLFS. Each row reports estimates of the treatment effects for a given year, specifically the coefficient of $Veiled_s * Post_t$ for veiled women, where $Post_t$ indicates the corresponding year. All regressions control for whether the woman is a native, dummies for each education level, age, household size, and region and year fixed effects, with their interactions. Standard errors are clustered at the individual level. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

Table A5: *Effects on Occupations and Work Conditions*

Outcome	Informal employment (1)	Permanent job (2)	Professionals (3)	Technicians (4)	Clerks (5)	Working hours (weekly) (6)	Earnings (hourly) (7)
Panel A							
Post	-0.013*** (0.001)	0.020*** (0.001)	0.000 (0.000)	-0.005*** (0.000)	-0.003*** (0.000)	0.519*** (0.050)	0.910*** (0.037)
Year fixed effects	No	No	No	No	No	No	No
Observations	934,060	934,060	934,060	934,060	934,060	917,988	165,666
Mean dependent var	0.15	0.18	0.03	0.01	0.01	12.73	9.85
Panel B							
Veiled $\times$ Post	-0.006*** (0.002)	0.018*** (0.002)	0.002** (0.001)	0.008*** (0.000)	0.003*** (0.000)	0.422*** (0.109)	-0.136* (0.070)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	934,060	934,060	934,060	934,060	934,060	917,988	165,666
Mean dependent var	0.18	0.11	0.01	0.01	0.01	10.69	7.31

Notes: Data are from the 2010-2017 HLFs. The sample includes all women aged 18-49. All regressions control for whether the woman is a native, dummies for each education level, age, household size, and region and year fixed effects, with their interactions. Standard errors are clustered at the individual level. Data are weighted using cross-sectional weights from the wave in which the outcome was measured. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

Table A6: *Effects on The Labor Market Outcomes of Women*

Outcome	Employment probability (1)	Public sector employment (2)	Private sector employment (3)	Self-employed (4)	Unpaid family worker (5)
Veiled $\times$ Post	0.011*** (0.002)	0.015*** (0.001)	0.001 (0.002)	-0.000 (0.001)	-0.011*** (0.001)
Post	0.070*** (0.006)	-0.030*** (0.003)	0.085*** (0.006)	0.023*** (0.002)	0.007*** (0.001)
Year fixed effects	No	No	No	No	No
Observations	934,060	934,060	934,060	934,060	934,060
Mean dependent var.	0.237	0.024	0.104	0.043	0.034

Notes: Data are from the 2010-2017 HLFS. The sample includes all women aged 18-49. The last row presents the mean outcome for all veiled women before the policy. Regressions control for whether the woman is a native, dummies for each education level, age, household size, and region and year fixed effects, with their interactions. Standard errors are clustered at the individual level. Data are weighted using cross-sectional weights from the wave in which the outcome was measured; **, significant at the 5 percent level; *, significant at the 10 percent level.

Table A7: Sensitivity Check For Labor Market Outcomes

Outcome	Employment probability (1)	Public sector employment (2)	Private sector employment (3)	Self-employed (4)	Unpaid family worker (5)
Panel A: Excluding the 2013 wave					
Veiled $\times$ Post	0.009*** (0.002)	0.017*** (0.001)	-0.000 (0.002)	-0.000 (0.001)	-0.013*** (0.001)
Post	0.071*** (0.006)	-0.030*** (0.003)	0.084*** (0.006)	0.023*** (0.002)	0.008*** (0.002)
Panel B: Using 2008 DHS for prediction and imputation					
Veiled $\times$ Post	0.012*** (0.002)	0.016*** (0.001)	-0.001 (0.002)	-0.000 (0.001)	-0.008*** (0.001)
Post	0.071*** (0.006)	-0.028*** (0.003)	0.086*** (0.006)	0.022*** (0.002)	0.004*** (0.001)
Panel C: Controlling for removing the ban in universities					
Veiled $\times$ Post	0.011*** (0.002)	0.015*** (0.001)	0.001 (0.002)	-0.000 (0.001)	-0.011*** (0.001)
After1991	0.004 (0.003)	0.034*** (0.001)	-0.029*** (0.003)	0.008*** (0.001)	-0.005*** (0.002)
After1991 $\times$ Post	-0.000 (0.003)	-0.020*** (0.001)	0.011*** (0.003)	-0.000 (0.001)	0.006*** (0.002)
Panel D: Controlling for the 1997 education reform					
Veiled $\times$ Post	0.009*** (0.002)	0.014*** (0.001)	-0.000 (0.002)	-0.000 (0.001)	-0.010*** (0.001)
After1986	0.008*** (0.003)	-0.034*** (0.001)	0.027*** (0.002)	-0.000 (0.001)	0.008*** (0.001)
After1986 $\times$ Post	-0.017*** (0.002)	-0.003*** (0.001)	-0.018*** (0.002)	0.001 (0.001)	0.004*** (0.001)
Panel E: Alternative sample using traditional gender role attitudes					
Traditional $\times$ Post	0.002 (0.002)	0.005*** (0.001)	-0.001 (0.002)	-0.000 (0.001)	-0.004*** (0.001)
Post	0.074*** (0.006)	-0.024*** (0.002)	0.086*** (0.006)	0.021*** (0.002)	0.003** (0.001)
F test (p value)	0.005	0.000	0.434	0.794	0.000
Panel F: Alternative sample using information on praying					
Praying $\times$ Post	0.008*** (0.003)	0.004*** (0.001)	0.005** (0.002)	-0.001 (0.001)	-0.002* (0.001)
Post	0.071*** (0.006)	-0.024*** (0.003)	0.082*** (0.006)	0.023*** (0.002)	0.002 (0.002)
F test (p value)	0.435	0.000	0.135	0.794	0.725

Notes: Data are from the 2010-2017 HLFS. The sample includes all women aged 18-49. All regressions control for whether the woman is a native, dummies for each education level, age, household size, and region and year fixed effects, with their interactions. Standard errors are clustered at the individual level. Data are weighted using cross-sectional weights from the wave in which the outcome was measured. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

Table A8: *Results assuming different partial correlations between veiling and employment probability*

Outcome	Employment probability	Public sector employment	Private sector employment	Self- employed	Unpaid family worker
	(1)	(2)	(3)	(4)	(5)
$\rho_{Y,Veiled X} = 0.10$					
Veiled $\times$ Post	0.013*** (0.002)	0.015*** (0.001)	0.003 (0.002)	-0.004*** (0.001)	-0.010*** (0.001)
Observations	935,075	935,075	935,075	935,075	935,075
$\rho_{Y,Veiled X} = 0.05$					
Veiled $\times$ Post	0.012*** (0.002)	0.015*** (0.001)	0.004** (0.002)	-0.003*** (0.001)	-0.012*** (0.001)
Observations	935,403	935,403	935,403	935,403	935,403

Notes: Data are from the 2010-2017 HLFS. The sample includes all women aged 18-49. Regressions control for native status, education level dummies, age, household size, and region and year fixed effects, along with their interactions. Standard errors (in parentheses) are clustered at the individual level. All estimates are weighted using cross-sectional survey weights from the wave in which the outcome was measured. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

Table A9: *Event Study Estimates for The Effects of The Policy Change on The Marriage Market Outcomes of Women*

Outcome	Marriage probability (1)	Husband has a university degree (2)	Husband has a high school degree (3)	Husband is a wage earner (4)	Age gap (5)
Veiled $\times$ 2011	0.007* (0.004)	-0.005 (0.004)	0.004 (0.005)	0.012** (0.006)	-0.026 (0.051)
Veiled $\times$ 2012	0.031*** (0.004)	-0.002 (0.004)	0.000 (0.005)	0.005 (0.006)	0.133** (0.052)
Veiled $\times$ 2013	0.022*** (0.004)	-0.007* (0.004)	0.006 (0.005)	0.009* (0.006)	0.035 (0.052)
Veiled $\times$ 2014	-0.007* (0.004)	0.001 (0.004)	-0.000 (0.005)	0.019*** (0.006)	0.090* (0.052)
Veiled $\times$ 2015	0.029*** (0.004)	-0.001 (0.004)	-0.006 (0.005)	0.030*** (0.006)	0.100* (0.053)
Veiled $\times$ 2016	0.026*** (0.004)	-0.004 (0.004)	0.003 (0.005)	0.036*** (0.006)	0.178*** (0.052)
Veiled $\times$ 2017	0.023*** (0.004)	-0.001 (0.004)	0.008 (0.005)	0.035*** (0.006)	0.106** (0.052)
Observations	934,060	648,747	648,747	648,747	648,747
Mean dependent var.	0.78	0.08	0.20	0.81	4.24

Notes: Data are from the HLFS. Each row reports estimates of the treatment effects for a given year, specifically the coefficient of $Veiled_s * Post_t$ for veiled women, where $Post_t$ indicates the corresponding year. All regressions control for whether the woman is a native, dummies for each education level, age, household size, and region and year fixed effects, with their interactions. Standard errors are clustered at the individual level. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

Table A10: Sensitivity Check For Marriage Market Outcomes

Outcome	Marriage probability (1)	Husband has a university degree (2)	Husband has a high school degree (3)	Husband is a wage earner (4)	Age gap (5)
Panel A: Including year $\times$ veiled interactions for pre-trends					
Veiled $\times$ Post	0.014*** (0.002)			0.028*** (0.004)	0.116*** (0.028)
Post	0.010** (0.005)			0.043*** (0.008)	0.102 (0.068)
Panel B: Excluding the 2013 wave					
Veiled $\times$ Post	0.005*** (0.002)	0.001 (0.002)	0.000 (0.003)	0.015*** (0.002)	0.083*** (0.028)
Post	0.010** (0.005)	0.036*** (0.005)	0.034*** (0.007)	0.033*** (0.006)	0.109 (0.068)
Panel C: Using 2008 DHS for prediction and imputation					
Veiled $\times$ Post	0.003 (0.002)	-0.001 (0.002)	0.004 (0.003)	0.021*** (0.002)	0.067** (0.026)
Post	0.013** (0.005)	0.037*** (0.005)	0.033*** (0.007)	0.032*** (0.006)	0.143** (0.067)
Panel D: Controlling for removing the ban in universities					
Veiled $\times$ Post	-0.001 (0.002)	0.002 (0.002)	-0.001 (0.003)	0.023*** (0.003)	0.090*** (0.026)
After1991	-0.130*** (0.003)	0.013*** (0.004)	-0.054*** (0.007)	0.026*** (0.007)	2.038*** (0.054)
After1991 $\times$ Post	0.026*** (0.003)	-0.002 (0.004)	-0.009 (0.007)	-0.017*** (0.006)	-0.691*** (0.062)
Panel E: Controlling for the 1997 education reform					
Veiled $\times$ Post	0.004** (0.002)	0.002 (0.002)	-0.002 (0.003)	0.015*** (0.002)	0.144*** (0.026)
After1986	-0.008*** (0.003)	-0.026*** (0.003)	-0.012*** (0.004)	0.020*** (0.003)	1.384*** (0.026)
After1986 $\times$ Post	0.018*** (0.002)	0.015*** (0.002)	-0.023*** (0.004)	-0.024*** (0.003)	-0.394*** (0.032)
Panel F: Alternative sample using traditional gender role attitudes					
Traditional $\times$ Post	0.003* (0.002)				
Post	0.011** (0.005)				
F test (p value)	0.792				
Panel G: Alternative sample using information on praying					
Praying $\times$ Post	-0.001 (0.002)				
Post	0.014*** (0.005)				
F test (p value)	0.316				

Notes: Data are from the 2010-2017 HLFS. The sample includes all women aged 18-49. All regressions control for whether the woman is a native, dummies for each education level, age, household size, and region and year fixed effects, with their interactions. Standard errors are clustered at the individual level. Data are weighted using cross-sectional weights from the wave in which the outcome was measured. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

Table A11: *Effects of The Policy Change on Labor Supply Gap between Partners*

Outcome	Working hours of women (1)	Working hours of men (2)	Working hours diff. (3)	Hourly earnings of women (4)	Hourly earnings of men (5)
Panel A					
Post	0.837*** (0.055)	-0.195*** (0.063)	-1.015*** (0.081)	0.768*** (0.051)	0.745*** (0.025)
Year fixed effects	No	No	No	No	No
Observations	636,627	621,150	611,782	93,066	343,443
Mean dependent var.	11.01	43.56	32.44	11.06	9.93
Panel B					
Veiled × Post	0.632*** (0.126)	0.391*** (0.133)	-0.209 (0.180)	-0.062 (0.101)	0.094 (0.060)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	636,627	621,150	611,782	93,066	343,443
Mean dependent var.	9.644	43.20	33.44	7.622	8.745

Notes: Data are from the 2010-2017 HLFS. The sample includes all married women aged 18-49. All regression models include controls for whether the female individual is native, dummies for each education level, age, age of the partners, dummies for the partner’s education and employment status, and a dummy for whether the partner is native and household size. Standard errors are clustered at the individual level. Data are weighted using cross-sectional weights from the wave in which the outcome was measured. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

Table A12: *Effects on labor market outcomes using Machine Learning matching*

Outcome	Employment probability (1)	Public sector employment (2)	Private sector employment (3)	Self-employed (4)	Unpaid family worker (5)
Panel A: Ensemble SVM+RF					
Veiled $\times$ Post	0.018*** (0.002)	0.022*** (0.001)	0.005** (0.002)	-0.003*** (0.001)	-0.015*** (0.001)
Panel B: Support Vector Machines					
Veiled $\times$ Post	0.024*** (0.003)	0.024*** (0.001)	0.006*** (0.002)	-0.002** (0.001)	-0.013*** (0.001)
Panel C: Random Forest					
Veiled $\times$ Post	0.019*** (0.002)	0.024*** (0.001)	0.003 (0.002)	-0.003*** (0.001)	-0.014*** (0.001)
Panel D: K-Nearest Neighbor					
Veiled $\times$ Post	0.015*** (0.002)	0.022*** (0.001)	0.003 (0.002)	-0.003*** (0.001)	-0.015*** (0.001)
Panel E: Neural Network					
PrVeiled $\times$ Post	0.039*** (0.005)	0.054*** (0.003)	0.005 (0.004)	-0.008*** (0.002)	-0.036*** (0.002)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	935,476	935,476	935,476	935,476	935,476
Mean dependent var.	0.523	0.082	0.286	0.109	0.063

Notes: Data are from the 2010-2017 HLFS. The sample includes full sample of women. Panels A-D show estimates from Eq. (1), representing binary measures of the treatment effect, while Panel E show estimates from Eq.(2), representing a continuous measures of the treatment effect. The last row presents the mean outcome for veiled women before the policy change. Regressions include controls for nativity, education (categorical), age, household size, and region and year fixed effects, along with their interactions. Standard errors are clustered at the individual level. All estimates are weighted using cross-sectional weights for the wave in which the outcome was measured. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.

Table A13: *Effects on Marriage Market Outcomes using Machine Learning matching*

Outcome	Marriage probability (1)	Husband has a university degree (2)	Husband has a high school degree (3)	Husband is a wage earner (4)	Age gap (5)
Panel A: Ensemble SVM+RF					
Veiled $\times$ Post	-0.003 (0.002)	0.003 (0.002)	0.003 (0.003)	0.024*** (0.002)	0.124*** (0.027)
Panel B: Support Vector Machines					
Veiled $\times$ Post	-0.005** (0.002)	0.002 (0.003)	0.005 (0.003)	0.026*** (0.002)	0.094*** (0.027)
Panel C: Random Forest					
Veiled $\times$ Post	0.000 (0.002)	0.005* (0.002)	0.003 (0.003)	0.024*** (0.002)	0.153*** (0.027)
Panel D: K-Nearest Neighbor					
Veiled $\times$ Post	-0.001 (0.002)	0.005** (0.002)	0.002 (0.003)	0.021*** (0.002)	0.149*** (0.027)
Panel E: Neural Network					
PrVeiled $\times$ Post	0.006 (0.004)	0.012*** (0.005)	0.004 (0.005)	0.062*** (0.005)	0.379*** (0.054)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	935,476	649,458	649,458	649,458	649,458
Mean dependent var.	0.78	0.15	0.22	0.82	4.20

Notes: Data are from the 2010-2017 HLFS. Column 1 uses the full sample of women, while Columns 2–5 are restricted to currently married women. The last row presents the mean outcome for veiled and married women before the policy change. Regressions include controls for nativity, education (categorical), age, household size, and region and year fixed effects, along with their interactions. Standard errors are clustered at the individual level. All estimates are weighted using cross-sectional weights for the wave in which the outcome was measured. ***, significant at the 1 percent level; **, significant at the 5 percent level; *, significant at the 10 percent level.